

Cover Page

Standard: BAC5004-4 Rev: (A) 06-Oct-2006

WARNING - The Export Control or Intellectual Property language on this page takes precedence over any such language that may appear on subsequent pages of this document.

No license is required for the dissemination of the commercial information contained herein to foreign persons other than those from or in the terrorist supporting countries identified in the United States Export Administration Regulations (EAR) (15 CFR 730-774). It is the responsibility of the individual in control of this data to abide by U.S. export laws.

ECCN 9E991

Boeing Proprietary, Distribution limited to Boeing employees and others covered by applicable proprietary information or nondisclosure agreements.

Copyright © 2003, 2006 Boeing. All rights reserved.

Any reproduction of this document or portions thereof must also reproduce this page of the document.

***** PSDS GENERATED *****

1 SCOPE

This specification and BAC5004 establish the requirements for the installation and inspection of non-fluid tight, permanent-type, straight shank, one sided installation (OSI) bolts in aluminum, steel, and titanium, or combinations thereof, as specified on the applicable Engineering drawings.

WARNING

WARNINGS may be included throughout this specification. Do not take these WARNINGS to be all inclusive, nor to completely describe hazards or precautionary measures applicable to specific procedures or operating environments.

Non-Boeing personnel must refer to their employer's safety instructions for information concerning hazards which may occur during operations described in this specification.

2 CLASSIFICATION

Not applicable to this specification.

3 REFERENCES

The current issue of the following documents shall be considered a part of this specification to the extent herein indicated.

BAC5000	– Sealing, General
BAC5004	– Installation of Permanent Fasteners
BAC5008	– Application of Lubricants
BAC5440	– Hole Preparation, Machining, and Grinding of Steels
BAC5492	– Machining and Cutting Titanium
BAC5736	– Application of Chemical and Solvent Resistant Finishes
BAC5750	– Solvent Cleaning
BAC5882	– Application of Urethane Compatible Primer
BACB30YH	– Bolt, One Sided Installation, 95 KSI Shear, A286 Body, 100 Degree Flush Shear Head (Nominal and Oversize)
BACB30YJ	– Bolt, One Sided Installation, 95 KSI Shear, A286 Body, Protruding Shear Head (Nominal and Oversize)
BACB30YR	– Bolt, One Sided Installation, 112 KSI Shear, Nickel Base Alloy Body, 100 Degree Flush Shear Head (Nominal and Oversize)
BACB30YS	– Bolt, One Sided Installation, 112 KSI Shear, Nickel Base Alloy Body, Protruding Shear Head (Nominal and Oversize)
BACB31L	– Bolt, One Sided Installation, 95 KSI Shear, Titanium, 100 Degree Flush Shear Head (Nominal and Oversize)
BACB31M	– Bolt, One Sided Installation, 95 KSI Shear, Titanium, Protruding Shear Head (Nominal and Oversize)

ORIGINAL ISSUE:
26-FEB-2003

REV: (ORG) 26-FEB-2003 (A) 06-OCT-2006

Authorizing Signatures on File

**ONE SIDED INSTALLATION BOLTS –
INSTALLATION AND INSPECTION**

**BAC
5004-4**

BOEING PROCESS SPECIFICATION

PAGE 1 of 45

***** PSDS GENERATED *****

3 REFERENCES (Continued)

- BSS7023 – Manufacturing Facilities with Approved Operating Procedures for
Manufacturing Process Control Documents
- BSS7286 – Statistical Process Control of Designated Engineering Characteristics

4 CONTENTS

<u>Section</u>	<u>Subject</u>	<u>Page</u>
1	SCOPE	1
2	CLASSIFICATION	1
3	REFERENCES	1
4	CONTENTS	2
5	MATERIALS CONTROL	6
6	FACILITIES CONTROL	6
7	DEFINITIONS	6
8	MANUFACTURING CONTROL	6
8.1	BACK SIDE CLEARANCE	6
8.2	GENERAL INFORMATION	10
8.3	HOLES	10
8.4	GRIP LENGTH	13
8.5	USE OF WASHERS	13
8.6	FLUSHNESS	14
8.7	INSTALLATION	14
8.7.1	COMPONENTS OF OSI BOLT	14
8.7.2	USE OF INSTALLATION TOOLING	14
8.7.3	PLACING BOLTS IN HOLES	21
8.7.4	INSTALLING BOLTS	21
8.8	CORROSION PREVENTION	22
8.8.1	INTERNAL INSTALLATIONS WITH ALUMINUM	22
8.8.2	EXTERNAL INSTALLATIONS WITH ALUMINUM	22
8.9	REWORK	23
8.9.1	FASTENER REPLACEMENT	23
8.9.2	FASTENER REMOVAL	25

BAC**5004-4**

Page 2

***** PSDS GENERATED *****

4 CONTENTS (Continued)

<u>Section</u>	<u>Subject</u>	<u>Page</u>
9	MAINTENANCE CONTROL	32
10	QUALITY CONTROL	33
10.1	IN-PROCESS VERIFICATION-VARIABLE MEASUREMENTS	33
10.2	IN-PROCESS VERIFICATION – ATTRIBUTE MEASUREMENTS	34
10.3	END ITEM VERIFICATION – VARIABLE MEASUREMENTS	34
10.4	END ITEM VERIFICATION – ATTRIBUTE MEASUREMENTS	35
11	REQUIREMENTS	35
11.1	HOLES	35
11.2	GRIP LENGTH	35
11.3	INSPECTION AFTER INSTALLATION	35
11.3.1	HEAD SIDE INSPECTION	35
11.3.2	BACK SIDE INSPECTION	36
12	TEST METHODS	45
13	QUALIFICATION	45

LIST OF FIGURES

<u>Number</u>	<u>Title</u>	<u>Page</u>
FIGURE 1	BACKSIDE CLEARANCE	6
FIGURE 2	ALLOWABLE BACKSIDE SLOPE	10
FIGURE 3	LIMITS FOR COUNTERSINK ANGULARITY AND ECCENTRICITY AND HOLE ANGULARITY	11
FIGURE 4	FILLET RADIUS/CHAMFER REQUIREMENT	12
FIGURE 5	GRIP GAGE	13
FIGURE 6	COMPONENTS OF OSI BOLT	14
FIGURE 7	INSTALLATION TOOL ASSEMBLY MOTOR SPECIFICATIONS	16
FIGURE 8	INSTALLING FASTENERS INTO INTERFERENCE FIT HOLE WITH RIVET SET TOOLING	20
FIGURE 9	INSTALLING BOLTS	22
FIGURE 10	REMOVAL TOOL ASSEMBLY	28

BAC**5004-4**

Page 3

***** PSDS GENERATED *****

4 CONTENTS (Continued)**LIST OF FIGURES** (Continued)

<u>Number</u>	<u>Title</u>	<u>Page</u>
FIGURE 11	ADJUSTING TOOL FOR COREBOLT REMOVAL	29
FIGURE 12	REMOVAL DRILL MOTOR ASSEMBLY	29
FIGURE 13	ADJUSTMENT OF GAP IN REMOVAL TOOL	30
FIGURE 14	DRILLING OUT COREBOLT HEAD	30
FIGURE 15	RIVET SET	31
FIGURE 16	PILOTED END MILL FOR HEAD TO SHANK REMOVAL OF BODY	31
FIGURE 17	PUNCHING COREBOLT, SLEEVE, AND NUT OUT THE BACKSIDE	31
FIGURE 18	PILOTED END MILL CUTTING OF BODY IN COUNTERSINK	32
FIGURE 19	PILOTED RIVET SET FOR REMOVAL OF BODY	32
FIGURE 20	INSTRUMENT FOR CHECKING FLUSHNESS	38
FIGURE 21	PROPERLY INSTALLED FASTENER – COREBOLT FRACTURED IN TORSION	38
FIGURE 22	IMPROPERLY INSTALLED FASTENER – COREBOLT FRACTURED IN BENDING	39
FIGURE 23	COREBOLT STEM BREAKOFF AND BULB FORMATION DIMENSIONS	41
FIGURE 24	BACK SIDE BULB FORMATION–UNACCEPTABLE WORKMANSHIP	42
FIGURE 25	INSPECTION CRITERIA FOR SLEEVE EXTRUSION	43
FIGURE 26	LEAF GAGE	44
FIGURE 27	BACK SIDE BULB INSPECTION	44
FIGURE 28	BACK SIDE INSPECTION–BULB DIAMETER NO GO	45
FIGURE 29	BACK SIDE INSPECTION BULB DIAMETER GO	45

BAC**5004-4**

Page 4

***** PSDS GENERATED *****

4 CONTENTS (Continued)**LIST OF TABLES**

<u>Number</u>	<u>Title</u>	<u>Page</u>
TABLE I	MINIMUM BACKSIDE CLEARANCE VALUES FOR A286 AND NI718 BOLTS	7
TABLE II	MINIMUM BACKSIDE CLEARANCE VALUES FOR TI BOLTS	9
TABLE III	FILLET RADIUS/CHAMFER REQUIREMENT	11
TABLE IV	HOLE DIAMETER LIMITS FOR BACB30YH AND BACB30YJ BOLTS	12
TABLE V	HOLE DIAMETER LIMITS FOR BACB30YR AND BACB30YS BOLTS	12
TABLE VI	HOLE DIAMETER LIMITS FOR NOMINAL BACB31L AND BACB31M BOLTS	13
TABLE VII	INSTALLATION TOOL SPECIFICATION	15
TABLE VIII	INSTALLATION TOOL ASSEMBLIES	18
TABLE IX	INSTALLATION TOOL ASSEMBLIES	19
TABLE X	HOLE DIAMETER LIMITS FOR OVERSIZE BACB30YH AND BACB30YJ BOLTS	23
TABLE XI	HOLE DIAMETER LIMITS FOR OVERSIZE BACB30YR AND BACB30YS BOLTS	24
TABLE XII	HOLE DIAMETER LIMITS FOR OVERSIZE BACB31L AND BACB31M BOLTS	24
TABLE XIII	REMOVAL TOOL COMPONENT PART NUMBERS	28
TABLE XIV	INITIAL RELIABILITY REQUIREMENT (IRR)	33
TABLE XV	SAMPLE SIZE REQUIREMENTS	34
TABLE XVI	ACCEPTABLE INSTALLED DIMENSIONS	39
TABLE XVII	GAGE SELECTION FOR BACK SIDE INSPECTION	43

BAC**5004-4**

Page 5

***** PSDS GENERATED *****

5 MATERIALS CONTROL

Not applicable to this specification.

6 FACILITIES CONTROL

Not applicable to this specification.

7 DEFINITIONS

Not applicable to this specification.

8 MANUFACTURING CONTROL**WARNING**

This specification involves the use of chemical substances which are hazardous. Boeing personnel shall refer to the work area Hazard Communication Handbook for health effect and control measure information contained in the HazCom Info Sheets and Material Safety Data Sheets. For disposition of hazardous waste materials, consult site environmental engineers for proper disposal methods.

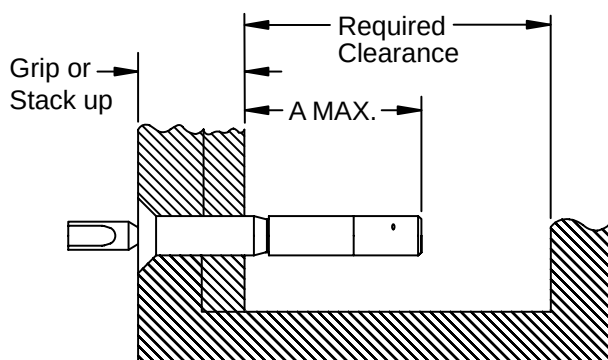
Non-Boeing personnel should refer to manufacturer's Material Safety Data Sheet(s) and their employer's safety instructions.

8.1

BACK SIDE CLEARANCE

Before installing the OSI bolt, consideration should be given to the minimum back side clearance. See Figure 1 and Table I. This is not an inspection requirement of Section 8 or Section 10.

NOTE: Values listed in Table I give the minimum back side clearance based on the space required to remove a discrepant installation by drilling to remove the corebolt head and pushing the corebolt through the body. Discrepant installations with insufficient backside clearance will require alternative removal methods not described by this specification.

**FIGURE 1 BACKSIDE CLEARANCE****BAC****5004-4**

Page 6

***** PSDS GENERATED *****

8.1 BACK SIDE CLEARANCE REQUIRED (Continued)**TABLE I MINIMUM BACKSIDE CLEARANCE VALUES FOR A286 AND NI718 BOLTS FL 1**

BACB30*	Bolt Diameter Code															Grip Dash Number FL 2
YH, YJ YR, YS	6() —	6()X —	— 8()	8() 8()X	8()X 8()Y	— 10()	10() 10()X	10()X 10()Y	— 12()	— 12()X	— 12()Y	— 14()	— 14()X	— 14()Y		
Amax	0.920	0.980	1.060	1.060	1.120	1.180	1.210	1.250	1.330	1.380	1.410	1.480	1.530	1.580		
P for PH	0.113	0.131	0.125	0.164	0.193	0.190	0.200	0.227	0.210	0.234	0.259	0.239	0.267	0.298	Protruding Head Minimum Back Side Clearance	
P for FH	0.000	0.018	−0.010	0.029	0.058	0.030	0.040	0.067	0.050	0.074	0.099	0.079	0.107	0.139		
	1.158	1.236	1.310	1.349	1.438	1.495	1.535	1.602	---	---	---	---	---	---		2
	1.220	1.298	1.372	1.411	1.500	1.557	1.597	1.664	1.727	1.801	1.856	1.906	1.984	2.065		3
	1.283	1.361	1.435	1.474	1.563	1.620	1.660	1.727	1.790	1.864	1.919	1.969	2.047	2.128		4
	1.345	1.423	1.497	1.536	1.625	1.682	1.722	1.789	1.852	1.926	1.981	2.031	2.109	2.190		5
	1.408	1.486	1.560	1.599	1.688	1.745	1.785	1.852	1.915	1.989	2.044	2.094	2.172	2.253		6
	1.470	1.548	1.622	1.661	1.750	1.807	1.847	1.914	1.944	2.051	2.106	2.156	2.234	2.315		7
	1.533	1.611	1.685	1.724	1.813	1.870	1.910	1.977	2.040	2.114	2.169	2.219	2.297	2.378		8
	1.595	1.673	1.747	1.786	1.875	1.932	1.972	2.039	2.102	2.176	2.231	2.281	2.359	2.440		9
	1.658	1.736	1.810	1.849	1.938	1.995	2.035	2.102	2.165	2.239	2.294	2.344	2.422	2.503		10
	1.720	1.798	1.872	1.911	2.000	2.057	2.097	2.164	2.227	2.301	2.356	2.406	2.484	2.565		11
	1.783	1.861	1.935	1.974	2.063	2.120	2.160	2.227	2.290	2.364	2.419	2.469	2.547	2.628		12
	1.845	1.923	1.997	2.036	2.125	2.182	2.222	2.289	2.352	2.426	2.481	2.531	2.609	2.690		13
	---	---	2.060	2.099	2.188	2.245	2.285	2.352	2.415	2.489	2.544	2.594	2.672	2.753		14
	---	---	2.122	2.161	2.250	2.307	2.347	2.414	2.477	2.551	2.606	2.656	2.734	2.815		15
	---	---	2.185	2.224	2.313	2.370	2.410	2.477	2.540	2.614	2.669	2.719	2.797	2.878		16
	---	---	---	---	---	2.432	2.472	2.539	2.602	2.676	2.731	2.781	2.859	2.940		17
	---	---	---	---	---	2.495	2.535	2.602	2.665	2.739	2.794	2.844	2.922	3.003		18
	---	---	---	---	---	---	---	---	2.727	2.801	2.856	2.906	2.984	3.065		19
	---	---	---	---	---	---	---	---	2.790	2.864	2.919	2.969	3.047	3.128		20
	---	---	---	---	---	---	---	---	---	---	---	3.031	3.109	3.190		21
	---	---	---	---	---	---	---	---	---	---	---	3.094	3.172	3.253	22	

BAC**5004-4**

Page 7

***** PSDS GENERATED *****

8.1

BACK SIDE CLEARANCE REQUIRED (Continued)**TABLE I MINIMUM BACKSIDE CLEARANCE VALUES FOR A286 AND
NI718 BOLTS** (Continued) **FL 1**

BACB30*	Bolt Diameter Code															Grip Dash Number FL 2
YH, YJ YR, YS	6() —	6()X —	— 8()	8() 8()X	8()X 8()Y	— 10()	10() 10()X	10()X 10()Y	— 12()	— 12()X	— 12()Y	— 14()	— 14()X	— 14()Y		
Amax	0.920	0.980	1.060	1.060	1.120	1.180	1.210	1.250	1.330	1.380	1.410	1.480	1.530	1.580		
P for PH	0.113	0.131	0.125	0.164	0.193	0.190	0.200	0.227	0.210	0.234	0.259	0.239	0.267	0.298		
P for FH	0.000	0.018	−0.010	0.029	0.058	0.030	0.040	0.067	0.050	0.074	0.098	0.079	0.107	0.139		
Flush Head Minimum Back Side Clearance	---	---	---	---	---	---	---	---	---	---	---	---	---	---	2	
	1.107	1.185	1.237	1.276	1.365	1.397	1.437	1.504	---	---	---	---	---	---	3	
	1.170	1.248	1.300	1.339	1.428	1.460	1.500	1.567	1.630	1.704	1.759	1.809	1.887	1.968	4	
	1.232	1.310	1.362	1.401	1.490	1.522	1.562	1.629	1.962	1.766	1.821	1.871	1.949	2.030	5	
	1.295	1.373	1.425	1.464	1.553	1.585	1.625	1.692	1.755	1.829	1.884	1.934	2.012	2.093	6	
	1.357	1.435	1.487	1.526	1.615	1.647	1.687	1.754	1.817	1.891	1.946	1.996	2.074	2.155	7	
	1.420	1.498	1.550	1.589	1.678	1.710	1.750	1.817	1.880	1.954	2.009	2.059	2.137	2.218	8	
	1.482	1.560	1.612	1.651	1.740	1.772	1.812	1.879	1.942	2.016	2.071	2.121	2.199	2.280	9	
	1.545	1.623	1.675	1.714	1.803	1.835	1.875	1.942	2.005	2.079	2.134	2.184	2.262	2.343	10	
	1.607	1.685	1.737	1.776	1.865	1.897	1.937	2.004	2.067	2.141	2.196	2.246	2.324	2.405	11	
	1.670	1.748	1.800	1.839	1.928	1.960	2.000	2.067	2.130	2.204	2.259	2.309	2.387	2.468	12	
	1.732	1.810	1.862	1.901	1.990	2.022	2.062	2.129	2.192	2.266	2.321	2.371	2.449	2.530	13	
	1.795	1.873	1.925	1.964	2.053	2.085	2.125	2.192	2.255	2.329	2.384	2.434	2.512	2.593	14	
	---	---	1.987	2.026	2.115	2.147	2.187	2.254	2.317	2.391	2.446	2.496	2.574	2.655	15	
	---	---	2.050	2.089	2.178	2.210	2.250	2.317	2.380	2.454	2.509	2.559	2.637	2.718	16	
	---	---	---	---	---	2.272	2.312	2.379	2.442	2.516	2.571	2.621	2.699	2.760	17	
	---	---	---	---	---	2.335	2.375	2.442	2.505	2.579	2.634	2.684	2.762	2.843	18	
	---	---	---	---	---	---	---	---	2.567	2.641	2.696	2.746	2.824	2.905	19	
	---	---	---	---	---	---	---	---	2.630	2.704	2.759	2.809	2.887	2.968	20	
	---	---	---	---	---	---	---	---	---	---	---	2.871	2.949	3.030	21	
	---	---	---	---	---	---	---	---	---	---	---	2.934	3.012	3.093	22	

FL 1 Back side protrusion spacing is based on the clearance required to remove a worst case discrepant installation (with limited back side access) by drilling to remove the manufactured head(s) and pushing the corebolt through the body. (See Figure 17 through Figure 19.)

FL 2 Minimum back side clearances required for grip dash numbers not listed are calculated by adding the fastener maximum grip length (noted on the applicable Boeing standard) to Amax (noted in Figure 1) and the constant P for the applicable head style. (Max Grip Length + Amax + P)

BAC**5004-4**

Page 8

***** PSDS GENERATED *****

8.1

BACK SIDE CLEARANCE REQUIRED (Continued)**TABLE II MINIMUM BACKSIDE CLEARANCE VALUES FOR TI BOLTS FL 1**

BACB31*	Bolt Diameter Code												Grip Dash Number FL 2
L, M	6()	6()X	6()Y	8()	8()X	8()Y	10()	10()X	10()Y	12()	12()X	12()Y	
Amax	0.870	0.870	0.930	1.050	1.050	1.100	1.100	1.130	1.170	1.270	1.320	1.350	2
P for PH	0.063	0.075	0.093	0.080	0.119	0.148	0.142	0.152	0.179	0.190	0.214	0.239	
P for FH	-0.012	0.000	0.018	-0.010	0.029	0.058	0.030	0.040	0.067	0.050	0.074	0.099	
Protruding Head Minimum Back Side Clearance	1.058	1.070	1.148	1.255	1.294	1.373	1.367	1.407	1.474	---	---	---	2
	1.120	1.132	1.210	1.317	1.356	1.435	1.429	1.469	1.536	1.647	1.721	1.776	3
	1.183	1.195	1.273	1.380	1.419	1.498	1.492	1.532	1.599	1.710	1.784	1.839	4
	1.245	1.257	1.335	1.442	1.481	1.560	1.554	1.594	1.661	1.772	1.846	1.901	5
	1.308	1.320	1.398	1.505	1.544	1.623	1.617	1.657	1.724	1.835	1.909	1.964	6
	1.370	1.382	1.460	1.567	1.606	1.685	1.679	1.719	1.786	1.897	1.971	2.026	7
	1.433	1.445	1.523	1.630	1.669	1.748	1.742	1.782	1.849	1.960	2.034	2.089	8
	1.495	1.507	1.585	1.692	1.731	1.810	1.804	1.844	1.911	2.022	2.096	2.151	9
	1.558	1.570	1.648	1.755	1.794	1.873	1.867	1.907	1.974	2.085	2.159	2.214	10
	1.620	1.632	1.710	1.817	1.856	1.935	1.929	1.969	2.036	2.147	2.221	2.276	11
	1.683	1.695	1.773	1.880	1.919	1.998	1.992	2.032	2.099	2.210	2.284	2.339	12
	1.745	1.757	1.835	1.942	1.981	2.060	2.054	2.094	2.161	2.272	2.346	2.401	13
	---	---	---	2.005	2.044	2.123	2.117	2.157	2.224	2.335	2.409	2.464	14
	---	---	---	2.067	2.106	2.185	2.179	2.219	2.286	2.397	2.471	2.526	15
	---	---	---	2.130	2.169	2.248	2.242	2.282	2.349	2.460	2.534	2.589	16
	---	---	---	---	---	---	2.304	2.344	2.411	2.522	2.596	2.651	17
	---	---	---	---	---	---	2.367	2.407	2.474	2.585	2.659	2.714	18
Flush Head Minimum Back Side Clearance	---	---	---	---	---	---	---	---	---	---	---	---	2
	1.045	1.057	1.135	1.227	1.266	1.345	1.317	1.357	1.424	---	---	---	3
	1.108	1.120	1.198	1.290	1.329	1.408	1.380	1.420	1.487	1.570	1.644	1.699	4
	1.170	1.182	1.260	1.352	1.391	1.470	1.442	1.482	1.549	1.632	1.706	1.761	5
	1.233	1.245	1.323	1.415	1.454	1.533	1.505	1.545	1.612	1.695	1.769	1.824	6
	1.295	1.307	1.385	1.477	1.516	1.595	1.567	1.607	1.674	1.757	1.831	1.886	7
	1.358	1.370	1.448	1.540	1.579	1.658	1.630	1.670	1.737	1.820	1.894	1.949	8
	1.420	1.432	1.510	1.602	1.641	1.720	1.692	1.732	1.799	1.882	1.956	2.011	9
	1.483	1.495	1.573	1.665	1.704	1.783	1.755	1.795	1.862	1.945	2.019	2.074	10
	1.545	1.557	1.635	1.727	1.766	1.845	1.817	1.857	1.924	2.007	2.081	2.136	11
	1.608	1.620	1.698	1.790	1.829	1.908	1.880	1.920	1.987	2.070	2.144	2.199	12
	1.670	1.682	1.760	1.852	1.891	1.970	1.942	1.982	2.049	2.132	2.206	2.261	13
	---	---	---	1.915	1.954	2.033	2.005	2.045	2.112	2.195	2.269	2.324	14
	---	---	---	1.977	2.016	2.095	2.067	2.107	2.174	2.257	2.331	2.386	15
	---	---	---	2.040	2.079	2.158	2.130	2.170	2.237	2.320	2.394	2.449	16
	---	---	---	---	---	---	2.192	2.232	2.299	2.382	2.456	2.511	17
	---	---	---	---	---	---	2.255	2.295	2.362	2.445	2.519	2.574	18

BAC**5004-4**

Page 9

***** PSDS GENERATED *****

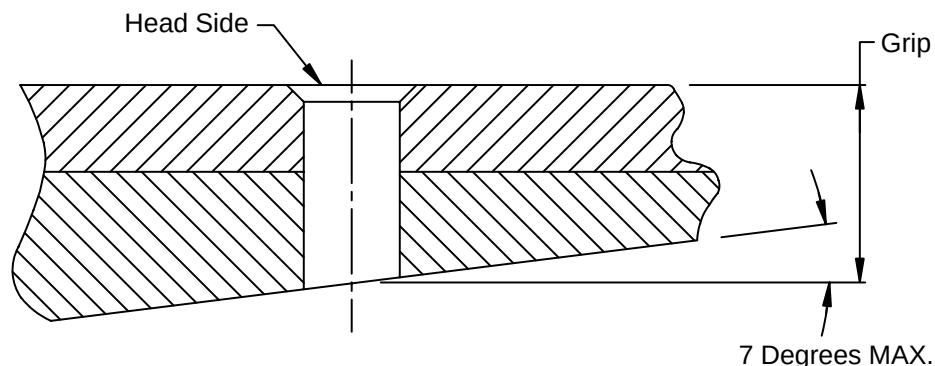
8.1 BACK SIDE CLEARANCE REQUIRED (Continued)**TABLE II MINIMUM BACKSIDE CLEARANCE VALUES FOR TI BOLTS** (Continued) **FL 1**

FL 1 Back side protrusion spacing is based on the clearance required to remove a worst case discrepant installation (with limited back side access) by drilling to remove the manufactured head(s) and pushing the corebolt through the body. (See Figure 17 through Figure 19.)

FL 2 Minimum back side clearances required for grip dash numbers not listed are calculated by adding the fastener maximum grip length (noted on the applicable Boeing standard) to Amax (noted in Figure 1) and the constant P for the applicable head style. (Max Grip Length + Amax + P)

8.2 GENERAL INFORMATION

- a. Only grip lengths appropriate to the sheet stackup should be selected. See Section 8.4.
- b. Only approved tools shall be used for installation or removal. See Table VIII, Table IX, and Table XIII.
- c. Sheets to be joined shall be firmly clamped and wedgelocks or equivalent used to help prevent hole misalignment.
- d. A slope of up to 7 degrees is allowable on the backside of the hole. See Figure 2.
- e. Bolts have been manufactured with a predetermined amount of lubrication required for performance. No addition or removal of lubricants in the installation process is allowed.

**FIGURE 2 ALLOWABLE BACKSIDE SLOPE**8.3 HOLES

- a. Prepare holes in accordance with BAC5004. Hole diameters for OSI bolts shall be in accordance with Table IV through Table VI unless otherwise specified on the drawing.
- b. Holes shall be drilled perpendicular to the sheet within 2 degrees maximum. Titanium and steel alloys shall be deburred in accordance with BAC5492 and BAC5440, respectively.

BAC**5004-4**

Page 10

***** PSDS GENERATED *****

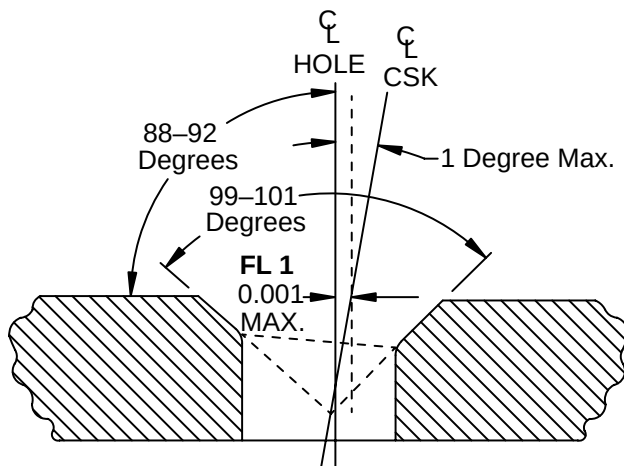
8.3

HOLES (Continued)

- c. Maximum hole angularity, countersink eccentricity, countersink included angle, and countersunk diameter for bolts installed in metal shall be controlled such that the installed bolt complies with the end item requirements shown in Figure 3 and Section 11 and BAC5004.
- d. All installations shall provide for bolt head to shank fillet relief. For protruding head bolt fillet relief, the hole shall have either a fillet radius or chamfer or a countersink washer shall be used with the bolt. For flush head bolts, the hole shall have a radius or chamfer at the junction of the hole and the countersink. See Figure 4 and Table III.

TABLE III FILLET RADIUS/CHAMFER REQUIREMENT

FASTENER DIAMETER SIZE	RADIUS OR CHAMFER
-6	0.030 to 0.040
-8	0.030 to 0.040
-10	0.040 to 0.050
-12	0.040 to 0.050
-14	0.050 to 0.060

**FIGURE 3 LIMITS FOR COUNTERSINK ANGULARITY AND ECCENTRICITY AND HOLE ANGULARITY**

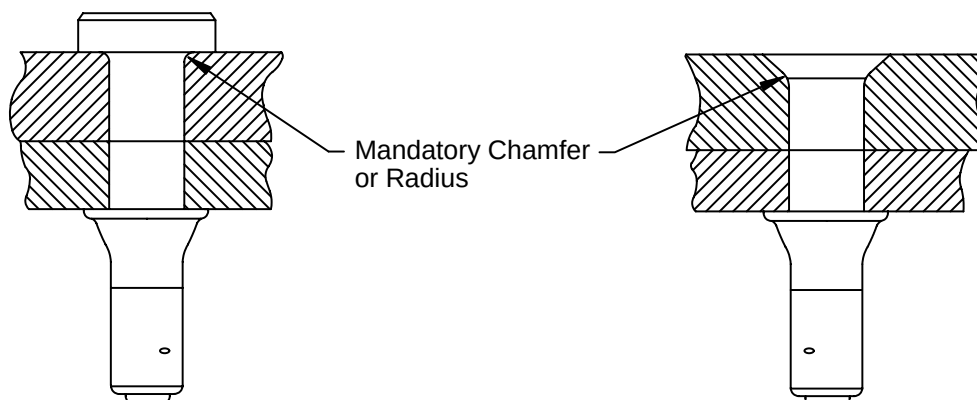
FL 1 Hole to countersink centerline eccentricity measured at surface of structure.

BAC**5004-4**

Page 11

***** PSDS GENERATED *****

8.3

HOLES (Continued)**FIGURE 4 FILLET RADIUS/CHAMFER REQUIREMENT****TABLE IV HOLE DIAMETER LIMITS FOR BACB30YH AND BACB30YJ BOLTS**

DASH NUMBER	HOLE DIAMETER LIMITS FOR ALUMINUM STRUCTURES	HOLE DIAMETER LIMITS FOR TITANIUM AND STEEL STRUCTURES AND COMBINATIONS
6()	0.2001 to 0.2021	0.2016 to 0.2026
8()	0.2626 to 0.2646	0.2641 to 0.2651
10()	0.3251 to 0.3271	0.3266 to 0.3276

TABLE V HOLE DIAMETER LIMITS FOR BACB30YR AND BACB30YS BOLTS FL 1

DASH NUMBER	HOLE DIAMETER LIMITS FOR TITANIUM AND STEEL STRUCTURES AND COMBINATIONS
8()	0.2485 to 0.2495
10()	0.3110 to 0.3120
12()	0.3735 to 0.3745
14()	0.4360 to 0.4370

FL 1 For aluminum structure applications, see BACB30YH and BACB30YJ.**BAC****5004-4**

Page 12

***** PSDS GENERATED *****

8.3 HOLES (Continued)

TABLE VI HOLE DIAMETER LIMITS FOR NOMINAL BACB31L AND BACB31M BOLTS

DASH NUMBER	HOLE DIAMETER LIMITS FOR TITANIUM AND STEEL STRUCTURES AND COMBINATIONS
6()	0.1885 to 0.1895
8()	0.2485 to 0.2495
10()	0.3110 to 0.3120
12()	0.3735 to 0.3745

8.4 GRIP LENGTH

- Grip lengths are given in .0625 increments. Available grip lengths are tabulated in BACB30YH, BACB30YJ, BACB30YR, BACB30YS, BACB31L, and BACB31M.
- Before bolt installation, grip lengths shall be verified for all holes with a standard grip gage and stack pulled up to remove gaps between sheets. See Figure 5 for illustration of grip gage.
- In those applications where a sloping sheet condition exists on the blind side, the grip length shall be determined by the depth at the centerline of the hole. This can be accomplished by first finding the minimum or maximum grip length and then rotating the grip length gage 90 degrees along the circumference of the hole. The grip length at this position will be equal to the grip length at the center of the hole.

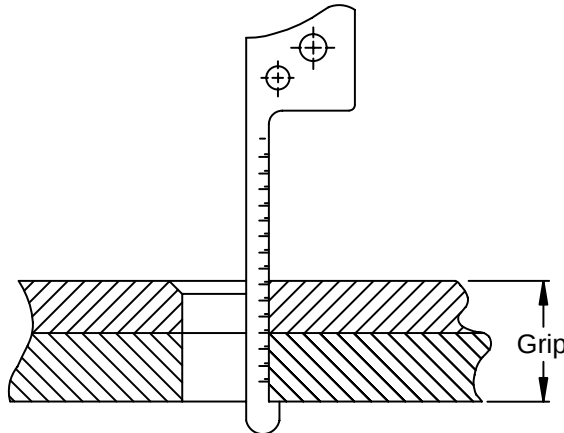


FIGURE 5 GRIP GAGE

8.5 USE OF WASHERS

Washers may be used for grip length adjustment with the following restrictions:

- No washers may be used on the bulb forming back side.

BAC**5004-4**

Page 13

***** PSDS GENERATED *****

8.5 USE OF WASHERS (Continued)

- b. Washers may be used for grip length adjustment on protruding head bolts only. If the correct grip length for a flush head bolt is not available, check with Engineering for an appropriate substitution.
- c. When washers are used for grip length adjustment, a countersink washer shall be used to provide head to shank fillet relief under the head.
- d. Unless otherwise specified, use a cadmium plated BACW10BP()CD washer against aluminum or alloy steel structure and a bare BACW10BP()ACU washer against titanium or corrosion resistant steel structure for fillet relief. Use a BACW10BP()DP or BACW30BP()APU washer, respectively, for grip adjustment.
- e. Only one grip adjusting washer may be used in addition to the fillet relief washer.

8.6 FLUSHNESS

- a. Flushness limits for flush-head bolts shall be $+0.010/-0.005$ inch unless otherwise noted on the engineering drawing. (See Figure 20.)
- b. Shaving of the stem breakoff area only to achieve flushness limits is allowed. Shaving of head of the body is not allowed. Shaving of more than the stem breakoff area on the protruding bolt head is not allowed. Shaving of back side protrusion is not allowed.

8.7 INSTALLATION

8.7.1 COMPONENTS OF OSI BOLT

The components of the OSI bolt are as shown in Figure 6.

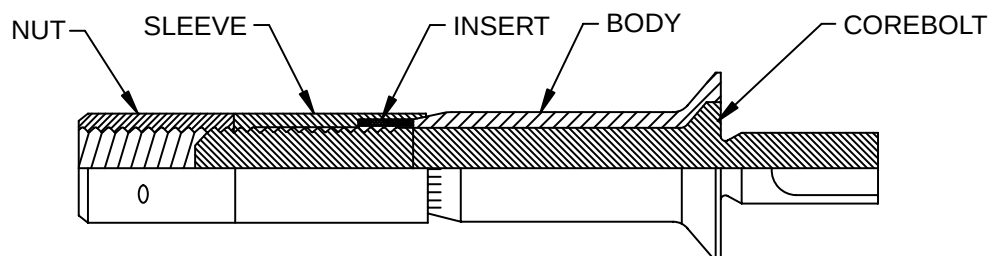


FIGURE 6 COMPONENTS OF OSI BOLT

8.7.2 USE OF INSTALLATION TOOLING

Only approved installation tooling shall be used meeting the minimum specifications indicated with an asterisk (*) in Table VII. See Figure 7 and Table VII, Table VIII, and Table IX for installation tool components.

BAC**5004-4**

Page 14

***** PSDS GENERATED *****

8.7.2 USE OF INSTALLATION TOOLING (Continued)

TABLE VII INSTALLATION TOOL SPECIFICATION

MOTOR SPECIFICATIONS	MOTORS FL 1						
	MP550OL	MRT300	MRL550	MP550OL	MP250OL	MP150OL	MRT300M
*Operating Pressure (psi recommended)	90	90	90	90	90	90	90
*Torque Output (ft-lbs @ 90 psi)	29.0	31.7	31.7	13.0	39.0	53.0	29.2
*Stall Torque (ft-lbs min)	19.9	32.0	31.0	26.4	76.0	105.6	32.0
*Motor Speed (rpm @ 90 psi/max)	400	300	300	400	135	130	300
Free Speed Air Consumption (cfm max)	26.0	22.0	22.0	26.0	32.0	34.0	32.0
Horsepower Rating	0.5	4.3	4.3	0.5	0.49	0.68	0.43
Weight (lbs)	3.5	4	4	3.5	3.5	7.5	4
Workspace Needed (in)	12	4	2.5				
Diameters	6-10	6-10	6-8	6-10	10-12	14	6-12
Availability	Obsolete			Current			

FL 1 Motors manufactured by Monogram Aerospace Fasteners, 3423 S. Garfield Ave.
Los Angeles, CA 98022-0547

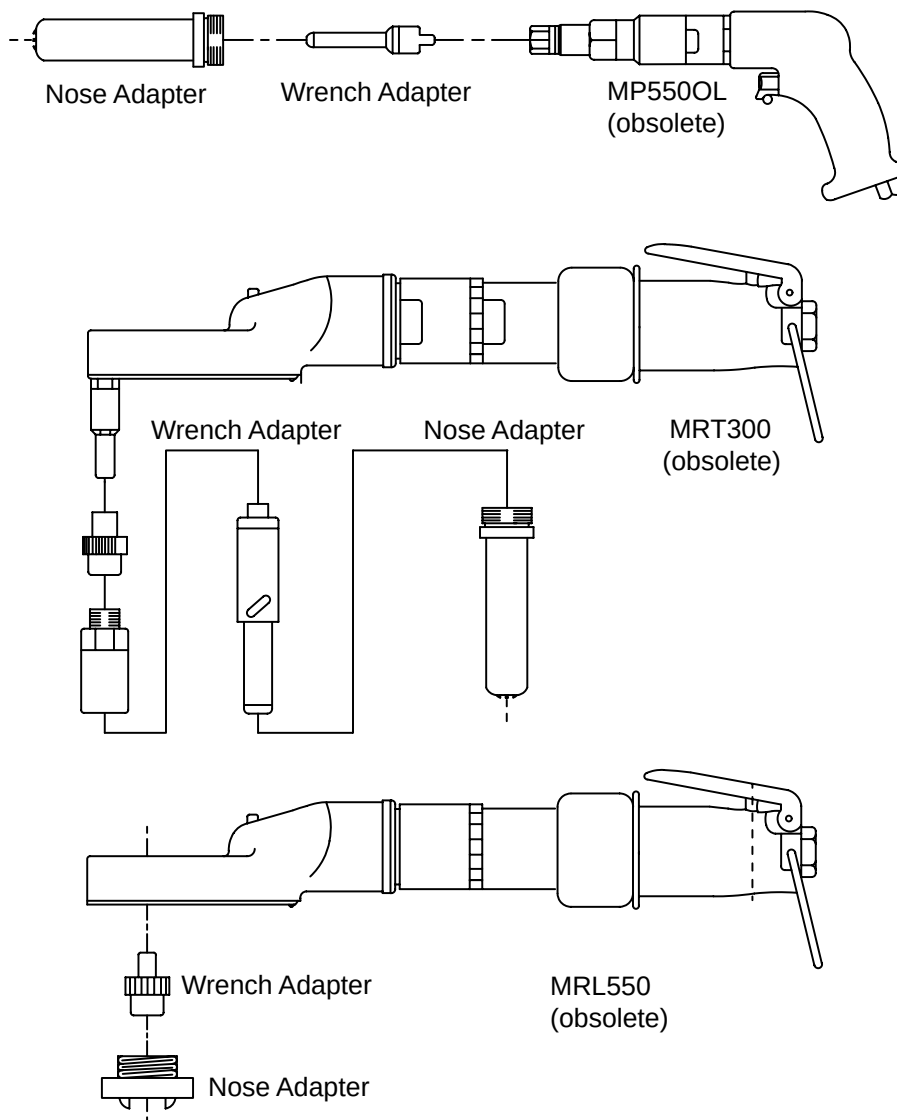
BAC**5004-4**

Page 15

***** PSDS GENERATED *****

8.7.2

USE OF INSTALLATION TOOLING (Continued)

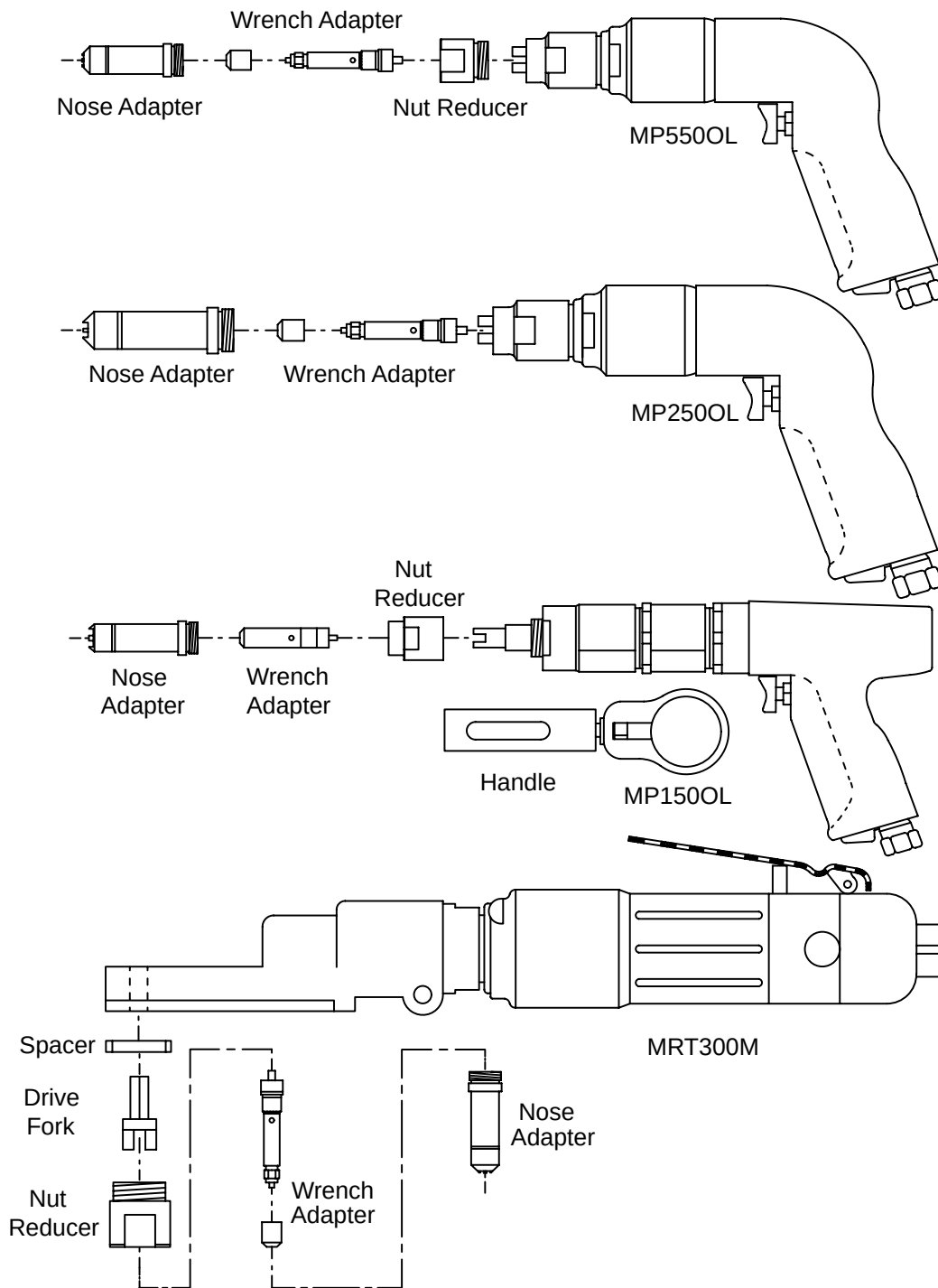
**FIGURE 7 INSTALLATION TOOL ASSEMBLY MOTOR SPECIFICATIONS****BAC****5004-4**

Page 16

***** PSDS GENERATED *****

8.7.2

USE OF INSTALLATION TOOLING (Continued)

**FIGURE 7 INSTALLATION TOOL ASSEMBLY MOTOR SPECIFICATIONS** (Continued)**BAC****5004-4**

Page 17

***** PSDS GENERATED *****

8.7.2

USE OF INSTALLATION TOOLING (Continued)

TABLE VIII INSTALLATION TOOL ASSEMBLIES FL 2

MOTOR	DIAMETER	WRENCH ADAPTER	NOSE ADAPTER	COMPLETE PNEUMATIC TOOL ASSEMBLY
MP550OL	-6()	MPBF-6	MPFOLS-6	MP550OLS-6AA
	-6()X			
	-6()Y			
	-8()	MPBF-8	MPFOS-8	MP550OLS-8AA
	-8()X			
	-8()Y			
	-10()	MPOL-10	MPFOLS-10	MP550OLS-10AA
	-10()X			
	-10()Y			
MRT300 (MRTHRL+ MRTG-1)	-6()	MTRTL-6	MRTFOL-6	MRT300OL-6AA
	-6()X			
	-6()Y			
	-8()	MTRTL-8	MRTFOL-8	MRT300OL-8AA
	-8()X			
	-8()Y			
	-10()	MTRTL-10	MRTFOL-10	MRT300OL-10AA
	-10()X			
	-10()Y			
MRL550	-6()	MRCBF-6	MRCFOL-6	MRC550OL-6AA
	-6()X			
	-6()Y			
	-8()	MRCBF-8	MRCFOL-8	MRC550OL-8AA
	-8()X			
	-8()Y			

FL 1 The tool assemblies listed herein are obsolete. See Table IX for current assemblies.

FL 2 Tools manufactured by Monogram Aerospace Fasteners, 3423 S. Garfield Ave. Los Angeles, CA 98022-0547

BAC**5004-4**

Page 18

***** PSDS GENERATED *****

8.7.2

USE OF INSTALLATION TOOLING (Continued)

TABLE IX INSTALLATION TOOL ASSEMBLIES FL 2

MOTOR	DIAMETER	CAP FL 1	WRENCH SUBADAPTER	NOSE ADAPTER	OTHER COMPONENTS	COMPLETE PNEUMATIC TOOL ASSY
MP550OL	-6()	MPCOL-06	MPWOL-01	MPFOL-06	MN550 NUT REDUCER	MP550OL-06AA
	-6()X			MPFOL-06-Y		MP550OL-06AAY
	-6()Y			MPFOL-08		MP550OL-08AA
	-8()	MPCOL-08		MPFOL-08-Y		MP550OL-08AAY
	-8()X					
	-8()Y					
	-10()	MPCOL-10	MPWOL-02	MPFOL-10	NONE	MP550OL-10AA
	-10()X			MPFOL-10-X		MP550OL-10AAX
	-10()Y			MPFOL-10-Y		MP550OL-10AAY
MP250OL	-10()	MPCOL-10	MPWOL-02	MPFOL-10	NONE	MP250OL-10AA
	-10()X			MPFOL-10X		MP250OL-10AAX
	-10()Y			MPFOL-10-Y		MP250OL-10AAY
	-12()	MPCOL-12		MPFOL-12		MP250OL-10AA
	-12()X			MPFOL-12-XY		MP250OL-10AAX
	-12()Y					
MP150OL	-14()	NONE	MPOL-14	MPFOL-14	NUT REDUCER, HP150L HANDLE	MP150OL-14AA
	-14()X			MPFOL-14X		MP150OL-14AAX
	-14()Y			MPFOL-14-Y		MP150OL-14AAY
MRT300 (MRT300H Motor + MRT300H Head)	-6()	MPCOL-06	MPWOL-01	MPFOL-06	MN550 NUT REDUCER, MDFOL DRIVE FORK, MNS300 SPACER	MRT300OL-06AA
	-6()X			MPFOL-06-Y		MRT300OL-06AAY
	-6()Y			MPFOL-08		MRT300OL-08AA
	-8()	MPCOL-08		MPFOL-08-Y		MRT300OL-08AAY
	-8()X			MPFOL-10		MRT300OL-10AA
	-8()Y			MPFOL-10-X		MRT300OL-10AAX
	-10()	MPCOL-10	MPWOL-02	MPFOL-10-Y		MRT300OL-10AAY
	-10()X			MPFOL-12		MRT300OL-12AA
	-10()Y			MPFOL-12-X		MRT300OL-12AAX
	-12()	MPCOL-12		MPFOL-12-Y		MRT300OL-12AAY
	-12()X					
	-12()Y					

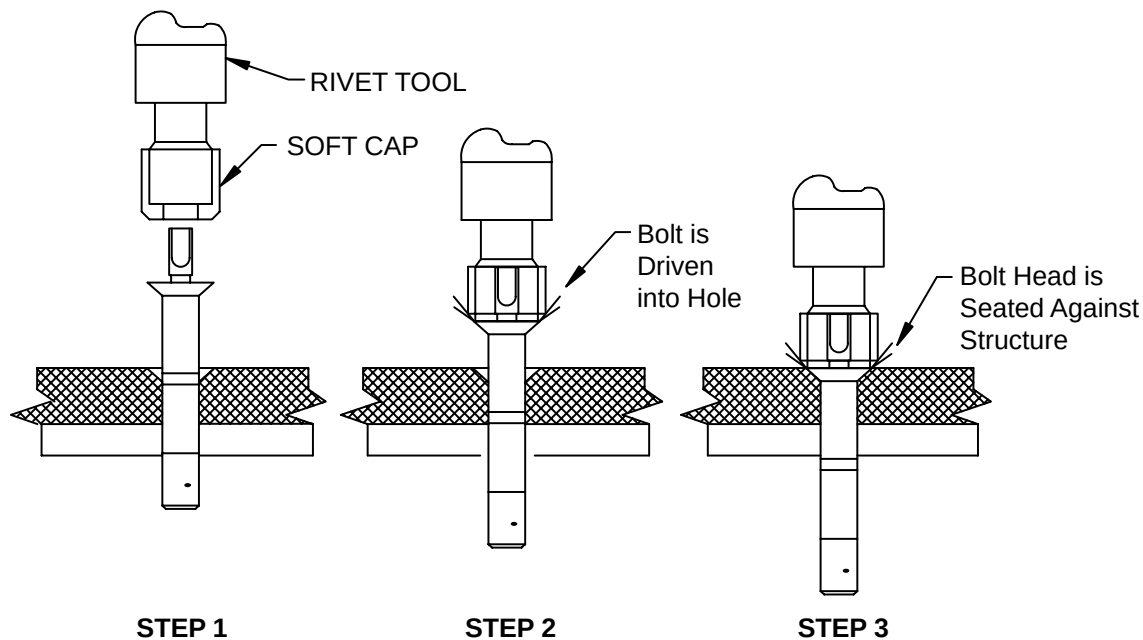
FL 1 Cap and wrench subadapter components form the complete wrench adapter.**FL 2** Tools manufactured by Monogram Aerospace Fasteners, 3423 S. Garfield Ave., Los Angeles, CA 98002-0547**BAC****5004-4**

Page 19

***** PSDS GENERATED *****

8.7.2

USE OF INSTALLATION TOOLING (Continued)



MONOGRAM INSTALLATION RIVET SET AND SOFT CAP PART NUMBER FL 1		
FASTENER DIAMETER	RIVET SET	SOFT CAP TOOL
-6	RC3330-06	RC3330-06-SC
-8	RC3330-08	RC3330-08-SC
-10	RC3330-10	RC3330-10-SC
-12	RC3330-12	RC3330-12-SC
-14	RC3330-14	RC3330-14-SC

FL 1 Tools manufactured by Monogram Aerospace Fasteners, 3423 S. Garfield Ave., Los Angeles, CA98022-0547

FIGURE 8 INSTALLING FASTENERS INTO INTERFERENCE FIT HOLE WITH RIVET SET TOOLING

BAC**5004-4**

Page 20

***** PSDS GENERATED *****

8.7.3 PLACING BOLTS IN HOLES

- a. Fasteners can be inserted into clearance holes with finger pressure.
- b. For interference holes, assistance from a hollow set rivet tool will be required. (See Figure 8 and Figure 15.) The protective "soft cap" should be placed over the nosepiece of the rivet tool which should then be placed over the breakoff stem so that it contacts the flush or protruding head surface. Drive the tool against the bolt into the panel until there is no gap between the bolt head and the panel. To insure good placement in the panel, care shall be taken to keep the rivet tool perpendicular to the bolt head surface and not allow angular loading. Do not apply force to the breakoff stem during insertion process.

8.7.4 INSTALLING BOLTS

Bolts shall be installed as shown in Figure 9. Only approved tools shall be used.

- a. The nose adapter of the driving tool is placed carefully over the corebolt breakoff stem so that the wrench adapter grips the corebolt wrench flats. The teeth at the end of the nose adapter are engaged in the recesses of the fastener body. The driving tool shall be held firmly, flush and perpendicular to the bolt so that no angular loading is induced.

CAUTION

Angular loading could cause bending, premature fracture of the breakoff stem, and an improper installation of the bolt.

- b. Power is steadily applied to the tool so that the wrench adapter rotates, turning the corebolt while the nose adapter holds the fastener body stationary. While the corebolt is turning, engagement of the corebolt thread through the nut will cause back side bulb formation of the sleeve. The stem is designed to shear off upon completion of the installation. If fracture of corebolt occurs during installation, inspect for potential structural damage.

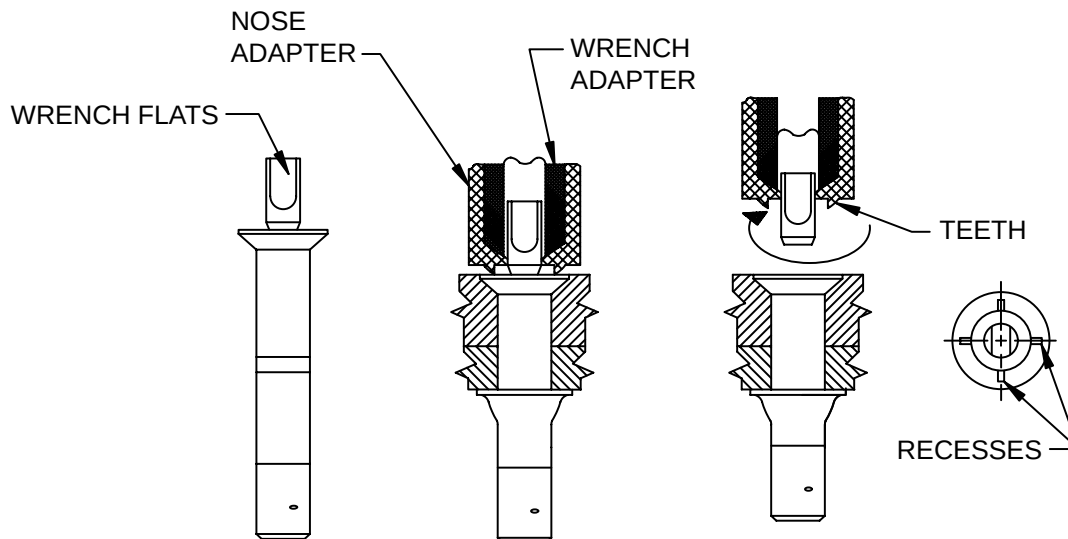
WARNING

In rare cases, corebolts have been known to fracture during installation resulting in a tail side projectile that can cause injury or damage to structure. Caution is advised to keep people clear on the back side from any possible shooting projectiles.

BAC**5004-4**

Page 21

***** PSDS GENERATED *****

8.7.4 INSTALLING BOLTS (Continued)**FIGURE 9 INSTALLING BOLTS**8.8 CORROSION PREVENTION

When installing a bolt where the sleeve forms against aluminum structure, a barrier coating of primer or sealant is required to prevent corrosion of the aluminum.

8.8.1 INTERNAL INSTALLATIONS WITH ALUMINUM

Before installation of the bolt in areas not exposed to the exterior environment, dip the nut and sleeve into BMS10-79 Type III, Class A, Grade A primer, prepared according to BAC5882, or into BMS10-11 Type I, Class A, Grade E primer, prepared according to BAC5736. Installation of bolts shall be done while primer is wet.

8.8.2 EXTERNAL INSTALLATIONS WITH ALUMINUM

Before installation of the bolt in areas that are exposed to the environment, apply BMS5-95 Class B or C, or BMS5-45 Class B or C sealant to the nut and sleeve in accordance with BAC5000, "Wet Installation of Fasteners" Method 3. Installation of bolts shall be done within the squeeze-out life of the sealant.

BAC**5004-4**

Page 22

***** PSDS GENERATED *****

8.9 REWORK

Document rework as required by the applicable quality assurance provisions.

8.9.1 FASTENER REPLACEMENT

If the fastener breaks during installation, spins after installation, or does not pass inspection after installation, it requires replacement. The two kinds of replacement are:

- a. Replacement with the same size fastener. This may be done if holes are not damaged (such as cracked, dimpled, out of round, scored or nicks) or out of tolerance, as verified by Quality Assurance.
- b. Replacement with oversize bolt. See Table X, Table XI and Table XII for oversize part numbers.

TABLE X HOLE DIAMETER LIMITS FOR OVERSIZE BACB30YH AND BACB30YJ BOLTS

ORIGINAL FASTENER DIAMETER CODE	OVERSIZE FASTENER DIAMETER CODE	HOLD DIAMETER LIMITS FOR ALUMINUM STRUCTURES		HOLE DIAMETER LIMITS FOR TITANIUM AND STEEL STRUCTURES AND COMBINATIONS	
		ORIGINAL	OVERSIZE FL 1	ORIGINAL	OVERSIZE FL 1
6()	6()X	0.2001 to 0.2021	0.2157 to 0.2177	0.2016 to 0.2026	0.2172 to 0.2182
8()	8()X	0.2626 to 0.2646	0.2782 to 0.2802	0.2641 to 0.2651	0.2797 to 0.2807
10()	10()X	0.3251 to 0.3271	0.3407 to 0.3427	0.3266 to 0.3276	0.3422 to 0.3432

FL 1 Engineering approval and Quality Assurance verification are required if edge margin is less than two diameters, whereas only Quality Assurance verification is required if the edge margin is twice the diameter of the original fastener, or greater.

BAC**5004-4**

Page 23

8.9.1

FASTENER REPLACEMENT (Continued)

TABLE XI HOLE DIAMETER LIMITS FOR OVERSIZE BACB30YR AND BACB30YS BOLTS

ORIGINAL FASTENER DIAMETER CODE	OVERSIZE FASTENER DIAMETER CODE	HOLE DIAMETER LIMITS FOR TITANIUM AND STEEL STRUCTURES AND COMBINATIONS	
		ORIGINAL	OVERSIZE FL 1
8()	8()X	0.2485 to 0.2495	0.2641 to 0.2651
8()X	8()Y	0.2641 to 0.2651	0.2797 to 0.2807
10()	10()X	0.3110 to 0.3120	0.3266 to 0.3276
10()X	10()Y	0.3266 to 0.3276	0.3422 to 0.3432
12()	12()X	0.3735 to 0.3745	0.3891 to 0.3901
12()X	12()Y	0.3891 to 0.3901	0.4047 to 0.4057
14()	14()X	0.4360 to 0.4370	0.4516 to 0.4526
14()X	14()Y	0.4516 to 0.4526	0.4672 to 0.4682

FL 1 Engineering approval and Quality Assurance verification are required if edge margin is less than two diameters, whereas only Quality Assurance verification is required if the edge margin is twice the diameter of the original fastener, or greater.

TABLE XII HOLE DIAMETER LIMITS FOR OVERSIZE BACB31L AND BACB31M BOLTS

ORIGINAL FASTENER DIAMETER CODE	OVERSIZE FASTENER DIAMETER CODE	HOLE DIAMETER LIMITS FOR TITANIUM AND STEEL STRUCTURES AND COMBINATIONS	
		ORIGINAL	OVERSIZE FL 1
6()	6()X	0.1885 to 0.1895	0.2016 to 0.2026
6()X	6()Y	0.2016 to 0.2026	0.2172 to 0.2182
8()	8()X	0.2485 to 0.2495	0.2641 to 0.2651
8()X	8()Y	0.2641 to 0.2651	0.2797 to 0.2807
10()	10()X	0.3110 to 0.3120	0.3266 to 0.3276
10()X	10()Y	0.3266 to 0.3276	0.3422 to 0.3432
12()	12()X	0.3735 to 0.3745	0.3891 to 0.3901
12()X	12()Y	0.3891 to 0.3901	0.4047 to 0.4057

FL 1 Engineering approval and Quality Assurance verification are required if edge margin is less than two diameters, whereas only Quality Assurance verification is required if the edge margin is twice the diameter of the original fastener, or greater.

BAC**5004-4**

Page 24

***** PSDS GENERATED *****

8.9.2

FASTENER REMOVAL

The two recommended methods of fastener removal are noted in items a. and b.:

- a. With easy back side access, the bolt can be cut through completely with an abrasive disk, or the corebolt can be broken by bending it at the base of the nut. Then, the remaining parts of the fastener are tapped through from the back or bulbed sleeve side out to the front (fastener head side) – using a stepped pin punch.

WARNING

Breaking or cutting the corebolt may result in a sudden release of components which can damage backside structure or injure shop personnel. Caution is advised to keep people clear during removal to avoid any possible shooting projectiles.

- b. When backside access is difficult, the corebolt head and portion of the body seated in the countersink may be drilled out and the remainder of the fastener assembly pushed through the back. Retrieve and remove remainder of fastener assembly from the backside. Refer to Table XIII for approved removal tools.

(1) Setting up the removal tool:

- (a) Insert appropriate diameter carbide star drill into air motor chuck until it bottoms out, then tighten chuck firmly. (See Figure 10.)
- (b) Slide appropriate nosepiece module over the drill bit and screw it onto the housing until the module bottoms out on the face of the housing (RC3075), then tighten.

(2) Setting cutting depth for corebolt removal:

- (a) Slide depth gage over nose adapter module. Using forefinger to apply slight downward pressure on the top of the depth gage, adjust cutting depth by rotating the housing (RC3075) until a gap is created between the base of the depth gage and the flange of the nose module. The gap is caused by the tip of the drill bit contacting the depth gage and pushing it upward. This procedure is to ensure that contact between the two components is made prior to facilitating step two of the depth control setup. (See Figure 11.)
- (b) The drill depth should not be set deeper than the bolt head height as indicated on the bolt standard (B dimension for flush head bolts, H dimension for protruding head bolts). After creating a gap and while continuing to apply slight forefinger pressure on the top of the depth gage, rotate the housing in the opposite direction until base of depth gage barely makes contact with the nosepiece flange. Hand tighten knurled lock ring (RC3101) up against the base of the housing to secure it and prevent further rotation. This process ensures that the housing will not rotate out of adjustment during the drilling process. Once the depth adjustment procedure is completed, the motor assembly should look like Figure 12.

BAC**5004-4**

Page 25

***** PSDS GENERATED *****

8.9.2

FASTENER REMOVAL (Continued)

(3) The drilling process:

- (a) The prongs at the tip of the nosepiece module shall first be engaged into the recesses of the one sided installation fastener. User should be careful to maintain motor assembly perpendicularity to the work surface, within 2 degrees. (See Figure 3.) This helps prevent drifting and aids in maintaining centerline cutting. Note that user shall bring motor to full RPM before making drill bit contact with the fastener or the carbide bit may chip and be rendered useless.
- (b) After reaching full RPM, it is recommended that the user peck away at the fastener corebolt as opposed to applying full drilling force immediately. Applying full force right away generally results in drifting off centerline of the fastener. The pecking technique is useful for creating a cone cavity, which acts as a makeshift guide when user eventually applies full force. Drilling should continue until the spring loaded gap as depicted in Figure 13 is closed, or the corebolt head becomes detached as depicted in Figure 14.

NOTE: Work hardening can occur in nickel and titanium alloy parts if positive cutting is not maintained.

- (c) The fixed travel of the nosepiece, in conjunction with the depth setting procedure, controls the cutting depth and should facilitate user drilling to the depth of the corebolt head. (See Figure 14).

(4) Corebolt Removal:

Once the corebolt head becomes detached, use the proper diameter rivet set (illustrated in Figure 13) and a hammer to tap the remaining corebolt, sleeve, and nut out the back side of the structure as shown in Figure 17.

WARNING

Use of pneumatic tools, such as rivet guns, to remove the corebolt may damage backside structure or injure shop personnel.

(5) Removing the remaining body component head in tight fit holes with end mill cutting method:

- (a) The piloted end mill shown in Figure 16 is to be utilized with any drill motor that does not exceed 200 RPM (optimum speed for cutting efficiency and end mill life).
- (b) Perpendicularity to work surface is to be maintained when cutting with an end mill. The depth of the cut shall not be set deeper than the fastener head height as indicated on the Boeing part standard. Full force should not be applied to end mill through drill motor. A light touch will likewise contribute to end mill life. Overcutting is not expected as the end mill pilot will insure cutting on the centerline.

BAC**5004-4**

Page 26

***** PSDS GENERATED *****

8.9.2

FASTENER REMOVAL (Continued)

(c) During cutting, liberal use of BoeLube in accordance with BAC5008 or equivalent is recommended. In most cases, the corebolt head will remain attached to the shank after the milling process. A very thin wall at this juncture is easily fractured when employing the piloted rivet set sequence shown in the next removal segment. However, depending on motor perpendicularity, the head may become detached during the milling action. See Figure 18.

(6) Removing the remaining shank portion of the OSI body:

(a) Do not use a standard hole punch to remove body. User shall select the appropriate size of piloted rivet set in accordance with Table XIII. This configuration varies from the standard installation rivet set in Figure 8 and Figure 15 in that a pilot on the end of the tool is employed to ensure that it remains confined within the OSI body. User should be careful to maintain the pilot perpendicular to the corebolt body head. See Figure 19.

NOTE: Be careful that the piloted rivet set has no burrs which could scrape the hole. Inspect hole for damage after removal.

(b) Pneumatic utility hammers may be used with the piloted rivet set. Power output of the pneumatic hammer shall be adjusted to a setting low enough to drive the fastener without damaging the surrounding structure. Test of the setting on a workbench before applying to the structure is recommended. Apply moderate end pressure as needed to drive out the body.

(7) Removing the remaining entire body component in clearance holes:

A temporary tool such as a cleco or Wedgelock may be used to pull the hollow body component through the hole from the installation side. The tool shall be sufficient to grasp the inner diameter or far end of the body component without the tool contacting and damaging the hole in the structure during removal.

BAC**5004-4**

Page 27

***** PSDS GENERATED *****

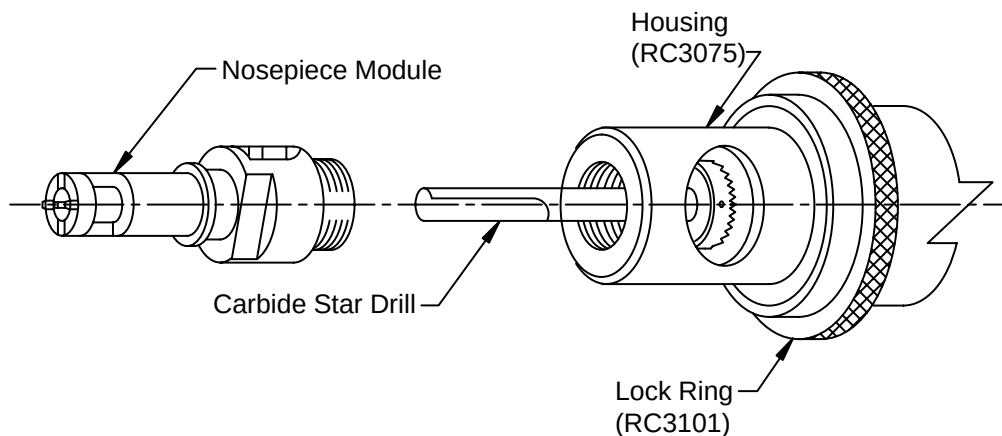
8.9.2 FASTENER REMOVAL (Continued)

TABLE XIII REMOVAL TOOL COMPONENT PART NUMBERS FL 1

BOLT SIZEBACB*			NOSE PIECE MODULE PART NUMBER		TOOL FOR COREBOLT REMOVAL		DEPTH GAGE		PILOTED TOOL FOR COMPONENT REMOVAL	
30YH	30YR	31L	FLUSH HEAD	PROTR. HEAD	CARBIDE STAR DRILL	RIVET SET	FLUSH HEAD	PROTR. HEAD	END MILL FOR HEAD TO SHANK	RIVET SET FOR BODY FL 2
30YJ	30YS	31M	RM3309	RM3311	RM3307	RM3331	RM3313	RM3313	RM3329	RM3328
---	---	6()	-06	-06	-06	-06	-01	-01	-06	-06
6()	---	6()X							-06X	-06X
6()X	---	6()Y							-06Y	-06Y
---	8()	8()	-08	-08	-08	-08	-01	-02	-08	-08
8()	8()X	8()X							-08X	-08X
8()X	8()Y	8()Y							-08Y	-08Y
---	10()	10()	-10	-10	-10	-10	-02	-02	-10	-10
10()	10()X	10()X							-10X	-10X
10()X	10()Y	10()Y							-10Y	-10Y
---	12()	12()	-12	-12	-12	-12	-02	-03	-12	-12
---	12()X	12()X							-12X	-12X
---	12()Y	12()Y							-12Y	-12Y
---	14()	---	-14	-14	-14	-14	-03	-04	-14	-14
---	14()X	---							-14X	-14X
---	14()Y	---							-14Y	-14Y
SEE FIGURE			12	12	14	15	11	11	18	19

FL 1 Gage manufactured by Monogram Aerospace Fasteners, 3423 S. Garfield Ave., Los Angeles, CA 90022-0547

FL 2 The shank of piloted end mills supplied by Monogram Aerospace Fasteners have been ground to 0.250 inch.

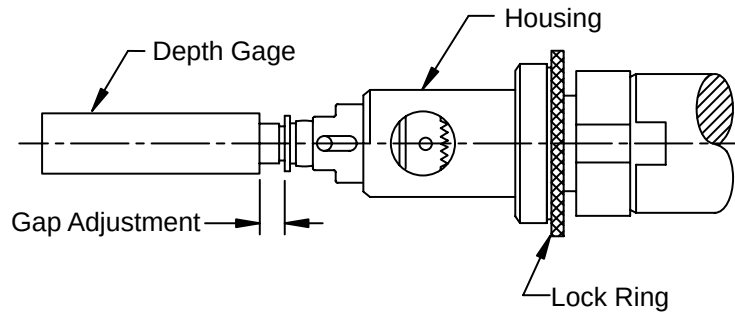
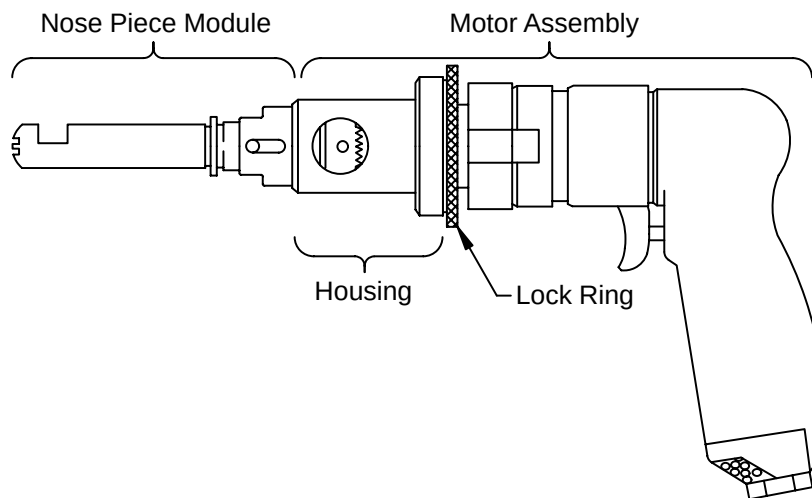
**FIGURE 10 REMOVAL TOOL ASSEMBLY****BAC****5004-4**

Page 28

***** PSDS GENERATED *****

8.9.2

FASTENER REMOVAL (Continued)

**FIGURE 11 ADJUSTING TOOL FOR COREBOLT REMOVAL****FIGURE 12 REMOVAL DRILL MOTOR ASSEMBLY****BAC****5004-4**

Page 29

***** PSDS GENERATED *****

■ 8.9.2

FASTENER REMOVAL (Continued)

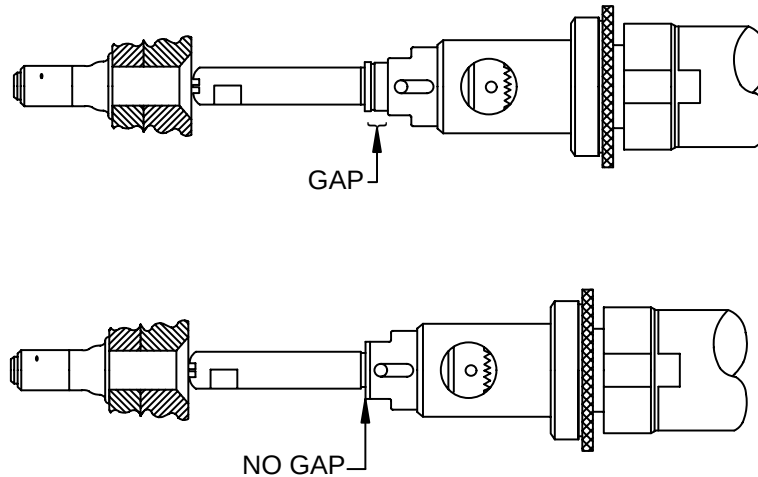


FIGURE 13 ADJUSTMENT OF GAP IN REMOVAL TOOL

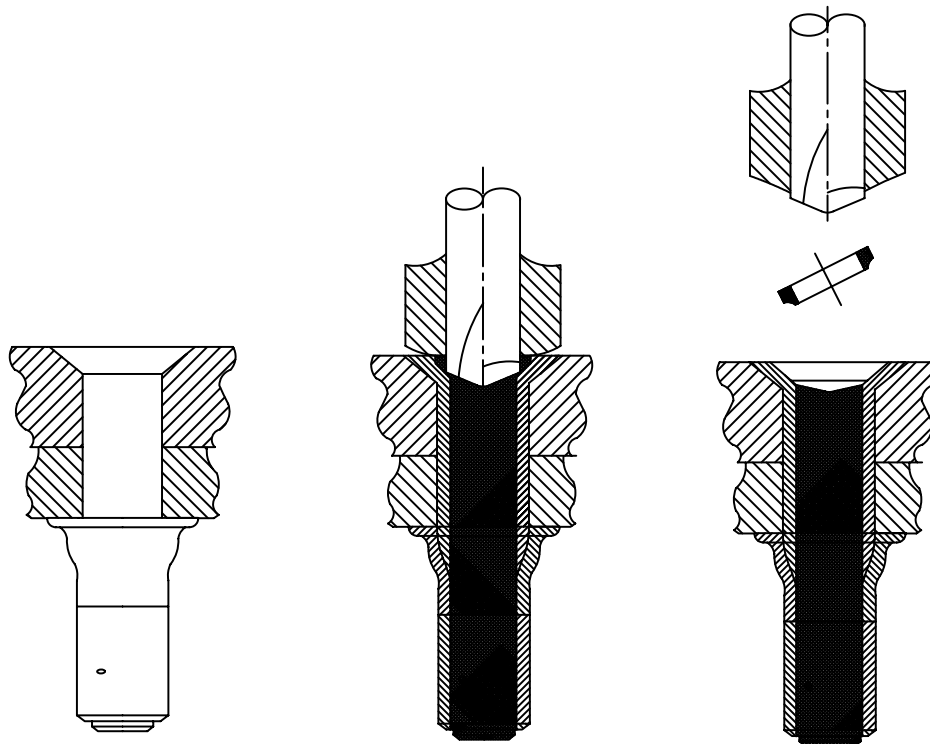


FIGURE 14 DRILLING OUT COREBOLT HEAD

BAC**5004-4**

Page 30

***** PSDS GENERATED *****

8.9.2

FASTENER REMOVAL (Continued)

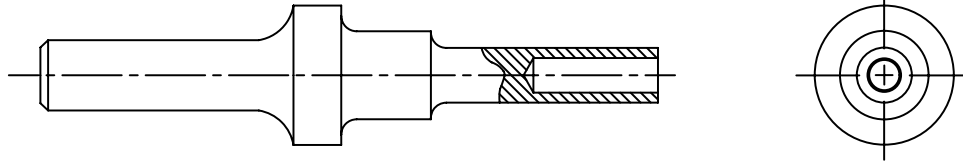


FIGURE 15 RIVET SET

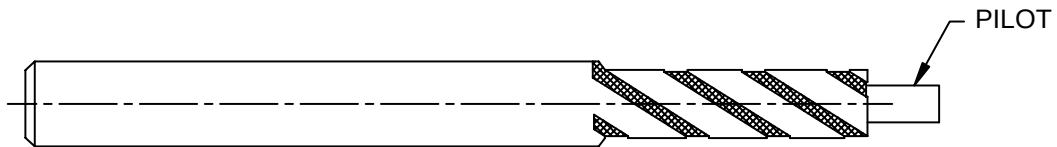


FIGURE 16 PILOTED END MILL FOR HEAD TO SHANK REMOVAL OF BODY

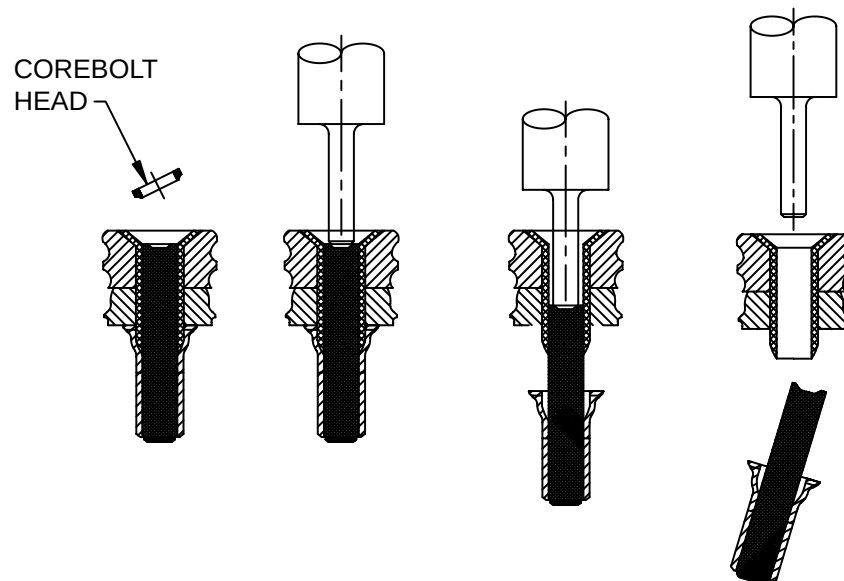


FIGURE 17 PUNCHING COREBOLT, SLEEVE, AND NUT OUT THE BACKSIDE

BAC**5004-4**

Page 31

***** PSDS GENERATED *****

8.9.2

FASTENER REMOVAL (Continued)

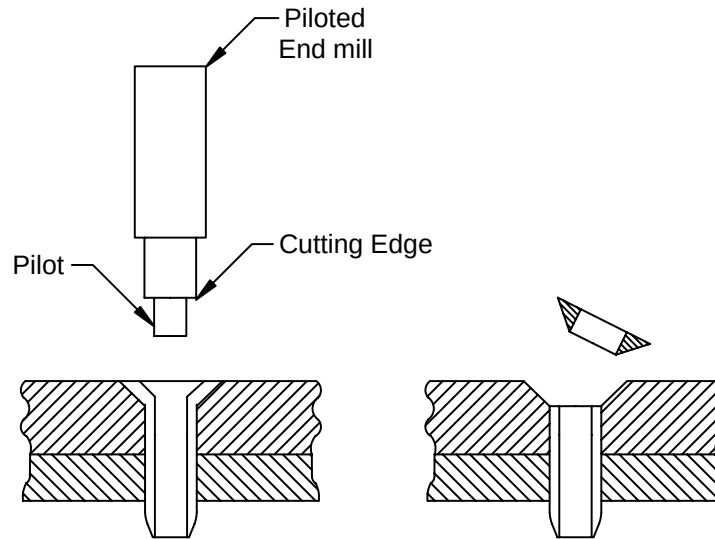


FIGURE 18 PILOTED END MILL CUTTING OF BODY IN COUNTERSINK

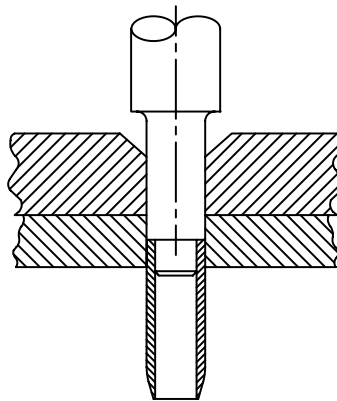


FIGURE 19 PILOTED RIVET SET FOR REMOVAL OF BODY

9

MAINTENANCE CONTROL

Verify that tools and gages used in this process are maintained and certified as required. Individual automated fastening systems shall be qualified in accordance with applicable, model specific Engineering criteria for each fastener type and variable.

BAC**5004-4**

Page 32

***** PSDS GENERATED *****

10

QUALITY CONTROL

- a. Assure that the requirements of this specification are met by monitoring the process and examining the end-items in accordance with established Quality Assurance provisions. The preferred method is to employ in-process monitoring using SPC analysis of variable measurement data, as described in Section 10.1 below. The other methods described in Sections 10.2 through 10.4 are acceptable alternatives. A combination or two or more methods may also be used. The sampling requirements contained in this document should not be construed as permitting acceptance of defective parts.
- b. Sampling plans shall meet the requirements in BSS7286 using the initial reliability requirement (IRR) value in Table XIV. IRR values stated in Table XIV define the level of verification required to demonstrate conformance.

TABLE XIV INITIAL RELIABILITY REQUIREMENT (IRR)

CHARACTERISTIC	IRR
1. Backside Visual Inspection	100 percent
2. Holes a. Diameter b. Surface defects c. Radius or chamfer d. Angularity e. Countersinking	98 percent
3. Flushness of Head	97 percent
4. Corebolt Stem Breakoff	
5. Gaps Under Backside Bulb	
6. Backside Bulb Formation	
7. All other Engineering Requirements	90 percent

10.1

IN-PROCESS VERIFICATION-VARIABLE MEASUREMENTS

- a. Each operator shall implement statistical process controls (SPC) at the start of a new process in accordance with BSS7286 for key characteristics and key process parameters. Once the documented results of the process indicate adequate process control, then the operator and quality assurance may agree to audit the process using other sampling and documentation methods. The sample statistic Cpk shall be at least 1.33 or yield at least 90 percent confidence that the production process is operating within the reliability requirements of Table XIV.
- b. If the operator uses SPC for any characteristic, and meets the requirements of BSS7286, then end-item sampling in accordance with Table XVI is not required for those characteristics. The appropriate sampling frequency for the SPC will cause a minimum of one applicable bolt assembly installation to be inspected for every eight made, as an average, and at least one per month regardless of rate.

BAC**5004-4**

Page 33

10.2

IN-PROCESS VERIFICATION – ATTRIBUTE MEASUREMENTS

- a. Quality Assurance shall inspect items in-process to insure the process is controlled in accordance with the approved operating procedure listed in BSS7023. Sample sizes are found in Table XV. If any units are found defective, then the entire lot is rejected.
- b. Lot definition and inspection procedures shall be in accordance with the operating division's Quality Assurance provisions.

TABLE XV SAMPLE SIZE REQUIREMENTS

LOT SIZE	INITIAL RELIABILITY REQUIREMENT				
	90	97	98	99	100
	SAMPLE SIZE				
Up to 25	6	All	All	All	All
26 to 50	6	19	25	All	All
51 to 102	6	20	29	50	All
103 to 106	6	21	30	51	All
107 to 111	6	21	30	52	All
112 to 116	6	21	30	53	All
117 to 123	6	21	30	54	All
124 to 130	6	21	31	55	All
131 to 140	6	21	31	56	All
141 to 155	6	21	31	57	All
156 to 200	6	22	32	58	All
201 to 214	7	22	32	59	All
215 to 233	7	22	32	60	All
234 to 270	7	22	32	61	All
271 to 323	7	22	33	62	All
324 to 403	7	22	33	63	All
404 to 502	7	22	33	64	All
503 to 633	7	22	33	65	All
634 to 917	7	22	34	66	All
918 to 1555	7	23	34	67	All
1556 to 5000	7	23	34	68	All
5001 and Up	7	23	34	69	All

10.3

END ITEM VERIFICATION – VARIABLE MEASUREMENTS

End item verification using variables measurements is acceptable for all characteristics which are end item inspectible.

BAC**5004-4**

Page 34

***** PSDS GENERATED *****

10.4 END ITEM VERIFICATION – ATTRIBUTE MEASUREMENTS

End item verification using inspection by attributes is acceptable for all characteristics which are end item inspectible.

11 REQUIREMENTS11.1 HOLES

Verify that holes conform to requirements of Sections 8.1, 8.2, and 8.3, or the engineering drawing.

11.2 GRIP LENGTH

Verify that all grip lengths conform to the requirements of Section 8.4. Unacceptable inspection results of Section 11.3.2 may be indications of improper grip length bolts.

11.3 INSPECTION AFTER INSTALLATION

100 percent visual inspection of OSI backside installation is required. Fastener leaf gage shall be used to verify bulb dimensions. See Table XIV and Section 11.3.2.

11.3.1 HEAD SIDE INSPECTION

- a. Head side shall meet corebolt stem breakoff limit requirements. See Section 10 and Table XIV for sampling requirements and see Table XVI for stem breakoff requirements.
- b. Figure 21 and Figure 22 are visual indications of proper or improper fastener installations respectively. In Figure 21, the texture of the breakoff stem is a shiny, torsional swirl, and in Figure 22, the texture is dark and grainy. Parts with breakneck areas as depicted in Figure 22 are acceptable provided all other requirements are met.
- c. Flushness And Corebolt Stem Breakoff
 - (1) A dial indicator with a tripod base or other base that straddles the fastener head (or otherwise rests on the adjacent surface) shall be used to check flushness. (Other measuring systems may also be used which are capable of measuring to -0.001 inch.) A contact point with no greater than 0.0312 inch wide tip shall be used. The contact point shall first be zeroed on the adjacent surface. Then the base shall be moved and the point brought to rest on the fastener body surface as shown. (Use of contact point with no greater than a 0.0312 inch wide tip will help prevent contact with corebolt head surface and body surface at the same time – which will cause an invalid measurement.) Bolts installed into surfaces composed of compound curves may require different verification techniques. See Figure 20.

CAUTION

Setting the pointer on the bolt's part mark can cause incorrect flushness readings.

BAC**5004-4**

Page 35

***** PSDS GENERATED *****

11.3.1 HEAD SIDE INSPECTION (Continued)

- (2) To measure for corebolt stem breakoff, equipment similar to that described in Section 11.3.1c.(1). shall be used, however, the contact point shall be zeroed on the fastener corebolt head surface and then brought to rest on the stem breakoff area. See Table XVI for stem breakoff limits.

d. Gap Evaluation

- (1) The heads of all flush and protruding head bolts shall be seated in at least one location on the periphery of the head. The head shall be seated such that a $0.002 + 0.0000 / - 0.0005$ inch thick shim cannot be inserted between the bolt head and the structure.
- (2) For measurements in curved surfaces, the heads of the flush fastener shall seat such that a $0.002 + 0.0000 / - 0.0005$ inch thick feeler gage pushed between the countersink and the head does not contact the shank and the feeler gage cannot be inserted between the head and the countersink for more than 33 percent of the circumference of the head.

11.3.2 BACK SIDE INSPECTION

a. Backside visual inspection:

Visual inspection sampling shall be in accordance with Table XIV and shall consist of the following:

- (1) Inspection shall be conducted from multiple viewpoints.
- (2) Workmanship requirements of Figure 24.
- (3) Visual evidence of one thread minimum protrusion from the top of the nut.
- (4) For bolts installed in structure where the blind side is in contact with aluminum, the nut and sleeve shall give visual evidence of having been coated with BMS10-11 or BMS10-79 primer.

b. Bulb formation:

For proper performance of the fastener, back side bulb shall meet the dimensional requirements of Figure 23 and Table XVI. A Leaf gage as depicted in Figure 26 shall be used to verify the installations. Sampling shall be in accordance with Table XIV. Additionally, any questionable bulbs shall be checked with a fastener leaf gage.

c. Gaging for inspection of back side bulb formation:

- (1) For each individual part number, a go-no go gage (shown in, Figure 26) is available to check whether the installed part is within the proper grip range and whether the bulb diameter is large enough. Refer to Table XVII to select the proper gage for the individual part number (for diameter and nominal size/oversize combination required).

BAC**5004-4**

Page 36

***** PSDS GENERATED *****

11.3.2

BACK SIDE INSPECTION (Continued)

(2) To check the grip condition, refer to Figure 27. Use the go-no go gage for the appropriate size fastener. For proper installation, the corebolt tip shall not touch the gage surface when placed as shown in Figure 27.

(3) If grip conditions are not met as indicated by the go-no go gage, the fastener is not properly installed. Quality assurance shall be contacted to document the discrepancy and engineering shall authorize all associated rework (See Section 8.9).

(4) To check the bulb condition, refer to Figure 28.

Using the appropriate gage from Table XVII, place the gage over the back side bulb as shown in Figure 28. If the gage clears the bulb diameter, as illustrated, the bulb is not large enough and the fastener is not properly installed (NO-GO).

A proper installation is shown in Figure 29. Here, the bulb is large enough where the gage will not clear it and sit on the structure surface (GO). A bulb diameter which meets the "go" condition is acceptable as long as it does not have any of the defects shown in Figure 24.

(5) If bulb diameter inspection indicates NO-GO, fastener is not properly installed. Quality assurance shall be contacted to document the discrepancy and engineering shall authorize all associated rework. (See Section 8.9.)

d. Corrosion protection:

For bolts installed in structure where the blind side is in contact with aluminum, the nut and sleeve shall give visual evidence of having been coated with BMS10-11 or BMS10-79 primer.

e. Thread protrusion:

Verify a minimum of one thread protrusion beyond top of nut.

f. Gap Evaluation

The backside bulb shall be seated such that a 0.002 +0.0000/-0.0005 inch thick feeler gage cannot be inserted between the bulb and the structure for more than 10 percent of the circumference of the bulb.

NOTE: Gapping under the bulb is more likely in installations in curved surfaces.

BAC**5004-4**

Page 37

***** PSDS GENERATED *****

11.3.2

BACK SIDE INSPECTION (Continued)

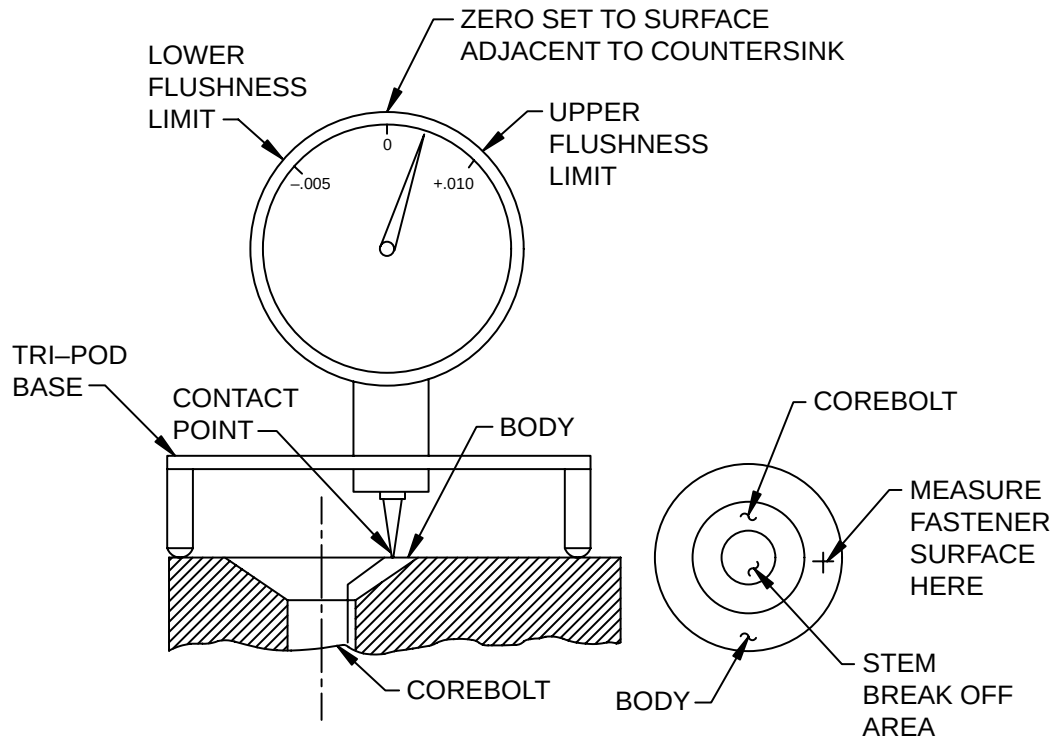


FIGURE 20 INSTRUMENT FOR CHECKING FLUSHNESS

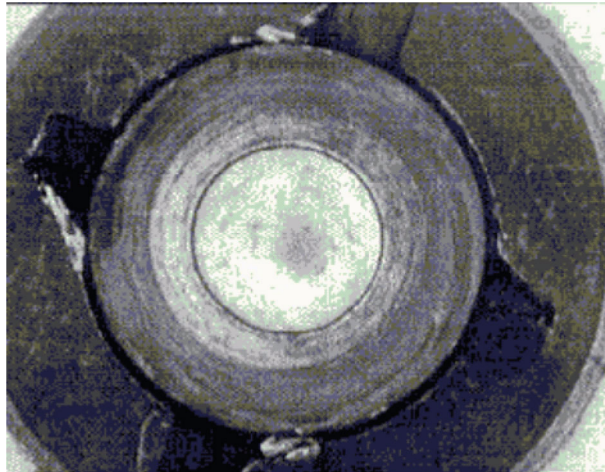


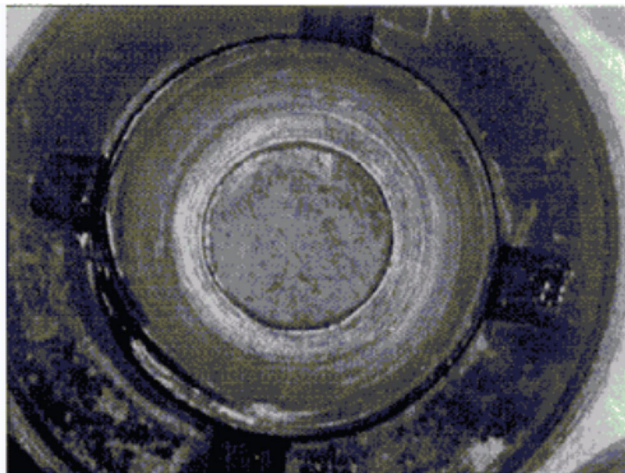
FIGURE 21 PROPERLY INSTALLED FASTENER – COREBOLT FRACTURED IN TORSION

BAC**5004-4**

Page 38

***** PSDS GENERATED *****

11.3.2 BACK SIDE INSPECTION (Continued)

**FIGURE 22 IMPROPERLY INSTALLED FASTENER – COREBOLT FRACTURED IN BENDING****TABLE XVI ACCEPTABLE INSTALLED DIMENSIONS**

FASTENER SIZE	BULB DIAMETER J MIN.	BULB DIAMETER K MAX.	STEM BREAKOFF X
BACB30YH AND BACB30YJ (95 KSI)			
6()	0.300	0.750	+ 0.015
6()X	0.320		– 0.005
8()	0.400	0.800	+ 0.020
8()X	0.400		– 0.005
10()	0.495	0.950	+ 0.020
10()X	0.515		– 0.005

BAC**5004-4**

Page 39

***** PSDS GENERATED *****

11.3.2

BACK SIDE INSPECTION (Continued)

TABLE XVI ACCEPTABLE INSTALLED DIMENSIONS (Continued)

FASTENER SIZE	BULB DIAMETER J MIN.	BACK SIDE PROTRUSION K MAX.	STEM BREAKOFF X
BACB30YR AND BACB30YS (112 KSI)			
8()	0.380	0.800	+ 0.020 – 0.005
8()X	0.400		
8()Y	0.400		
10()	0.475	0.950	+ 0.020 – 0.005
10()X	0.495		
10()Y	0.515		
12()	0.560	1.100	+ 0.020 – 0.005
12()X	0.580		
12()Y	0.600		
14()	0.660	1.250	+ 0.020 – 0.005
14()X	0.680		
14()Y	0.700		
BACB31L AND BACB31M (95 KSI)			
6()	0.290	0.750	+ 0.015 – 0.005
6()X	0.300		
6()Y	0.320		
8()	0.380	0.800	+ 0.020 – 0.005
8()X	0.400		
8()Y	0.400		
10()	0.475	0.950	+ 0.020 – 0.005
10()X	0.495		
10()Y	0.515		
12()	0.560	1.100	+ 0.020 – 0.005
12()X	0.580		
12()Y	0.600		

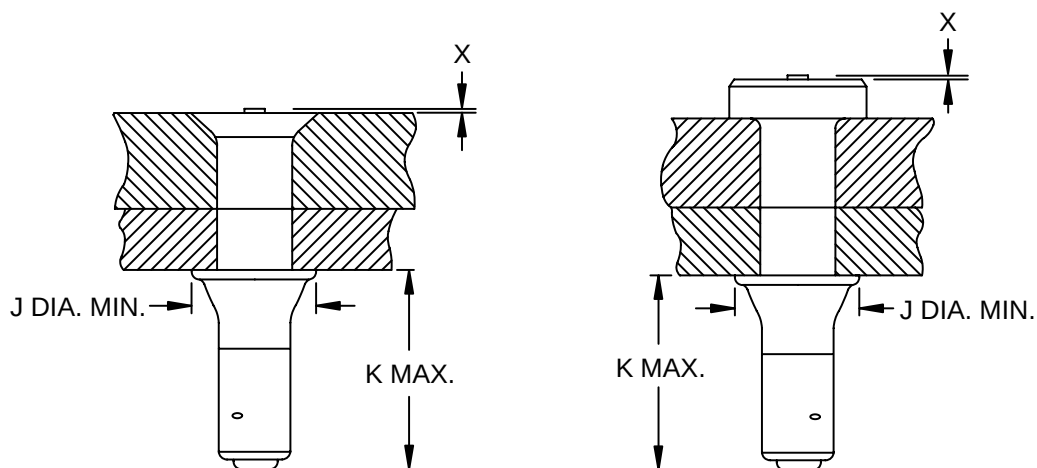
BAC**5004-4**

Page 40

***** PSDS GENERATED *****

11.3.2

BACK SIDE INSPECTION (Continued)



100 DEGREE FLUSH HEAD

PROTRUDING HEAD

FIGURE 23 COREBOLT STEM BREAKOFF AND BULB FORMATION DIMENSIONS

BAC**5004-4**

Page 41

***** PSDS GENERATED *****

11.3.2 BACK SIDE INSPECTION (Continued)

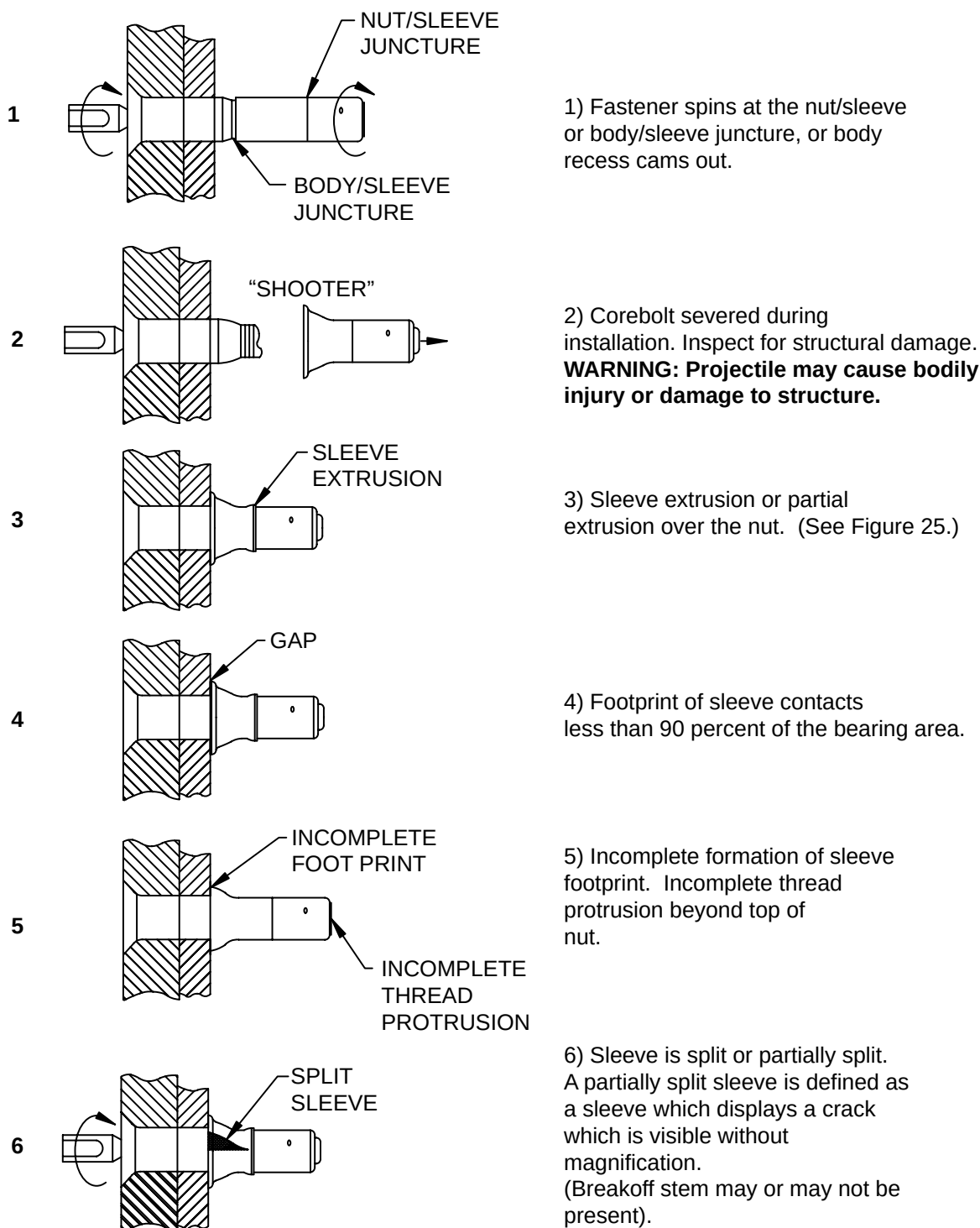
- 
- 1) Fastener spins at the nut/sleeve or body/sleeve juncture, or body recess cams out.
- 2) Corebolt severed during installation. Inspect for structural damage. **WARNING: Projectile may cause bodily injury or damage to structure.**
- 3) Sleeve extrusion or partial extrusion over the nut. (See Figure 25.)
- 4) Footprint of sleeve contacts less than 90 percent of the bearing area.
- 5) Incomplete formation of sleeve footprint. Incomplete thread protrusion beyond top of nut.
- 6) Sleeve is split or partially split. A partially split sleeve is defined as a sleeve which displays a crack which is visible without magnification. (Breakoff stem may or may not be present).

FIGURE 24 BACK SIDE BULB FORMATION—UNACCEPTABLE WORKMANSHIP

BAC

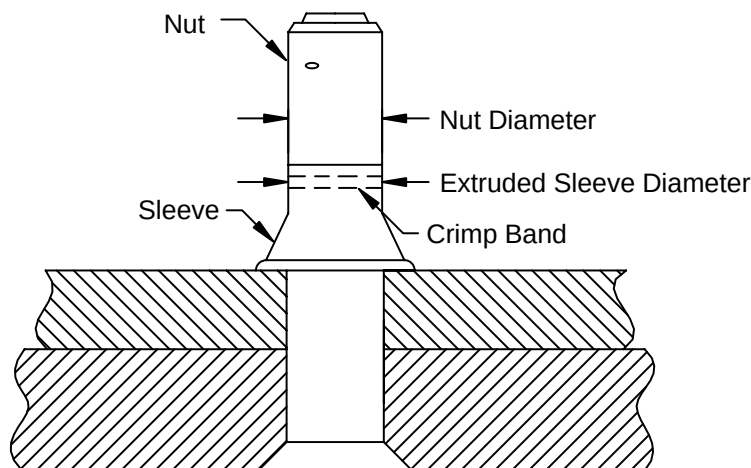
5004-4

Page 42

***** PSDS GENERATED *****

11.3.2

BACK SIDE INSPECTION (Continued)

**FIGURE 25 INSPECTION CRITERIA FOR SLEEVE EXTRUSION**

Sleeve diameter at center of crimp band (measured approximately 0.031 from the nut/sleeve juncture) shall not exceed the nut base diameter by more than 0.050.

TABLE XVII GAGE SELECTION FOR BACK SIDE INSPECTION

DIAMETER DASH NUMBER			LEAF GAGE PART NUMBER FL 1	GAGE DIMENSIONS	
BACB*				A + 0.000 − 0.002	B + 0.002 − 0.000
30YH	30YR	31L			
30YJ	30YS	31M			
---	---	6()	MGOSI-6	0.750	0.290
6()	---	6()X	MGOSI-6X		0.300
6()X	---	6()Y	MGOSI-6Y		0.320
---	8()	8()	MGOSI-8	0.800	0.380
8()	8()X	8()X	MGOSI-8XY		0.400
8()X	8()Y	8()Y			0.400
---	10()	10()	MGOSI-10	0.950	0.475
10()	10()X	10()X	MGOSI-10X		0.495
10()X	10()Y	10()Y	MGOSI-10Y		0.515
---	12()	12()	MGOSI-12	1.100	0.560
---	12()X	12()X	MGOSI-12X		0.580
---	12()Y	12()Y	MGOSI-12Y		0.600
---	14()	---	MGOSI-14	1.250	0.660
---	14()X	---	MGOSI-14Y		0.680
---	14()Y	---	MGOSI-14Y		0.700

FL 1 Gage manufactured by Monogram Aerospace Fasteners, 3423 S. Garfield Ave., Los Angeles, CA 98022-0547

BAC**5004-4**

Page 43

***** PSDS GENERATED *****

11.3.2

BACK SIDE INSPECTION (Continued)

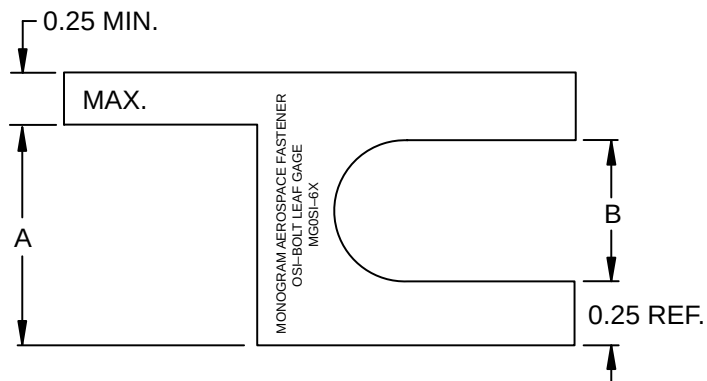


FIGURE 26 LEAF GAGE

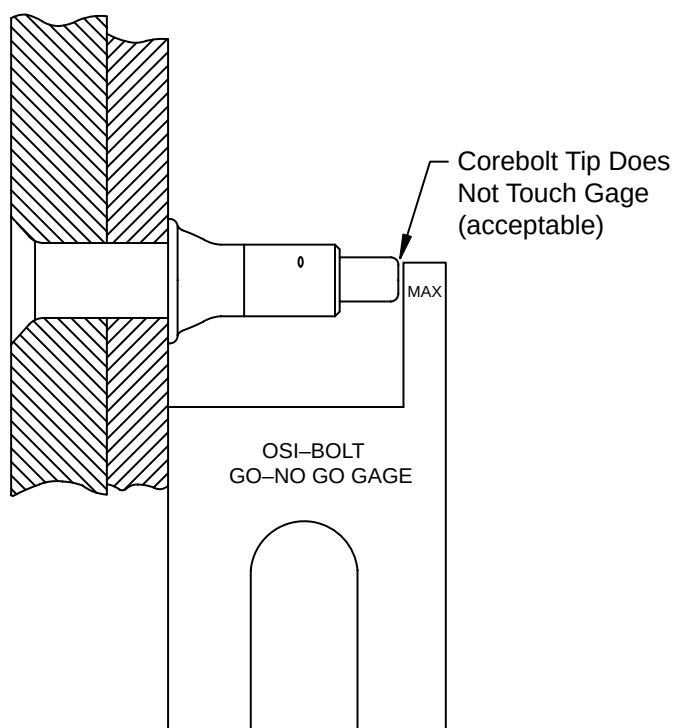


FIGURE 27 BACK SIDE BULB INSPECTION

BAC**5004-4**

Page 44

***** PSDS GENERATED *****

11.3.2

BACK SIDE INSPECTION (Continued)

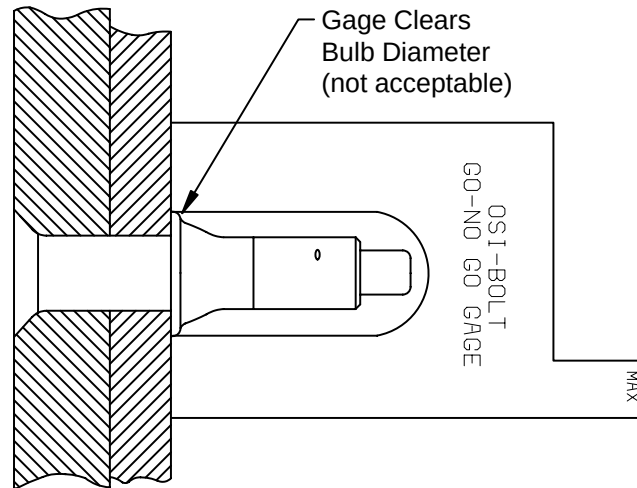


FIGURE 28 BACK SIDE INSPECTION-BULB DIAMETER NO GO

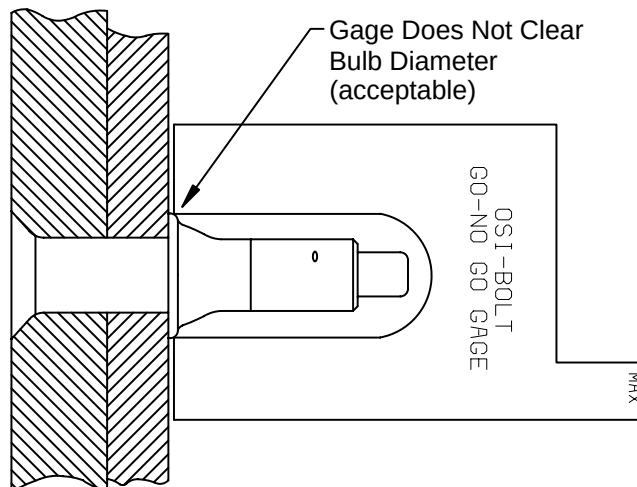


FIGURE 29 BACK SIDE INSPECTION BULB DIAMETER GO

12

TEST METHODS

Not applicable to this specification.

13

QUALIFICATION

Not applicable to this specification.

BAC**5004-4**

Page 45